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(54) POWERED TWO-WHEEL VEHICLE

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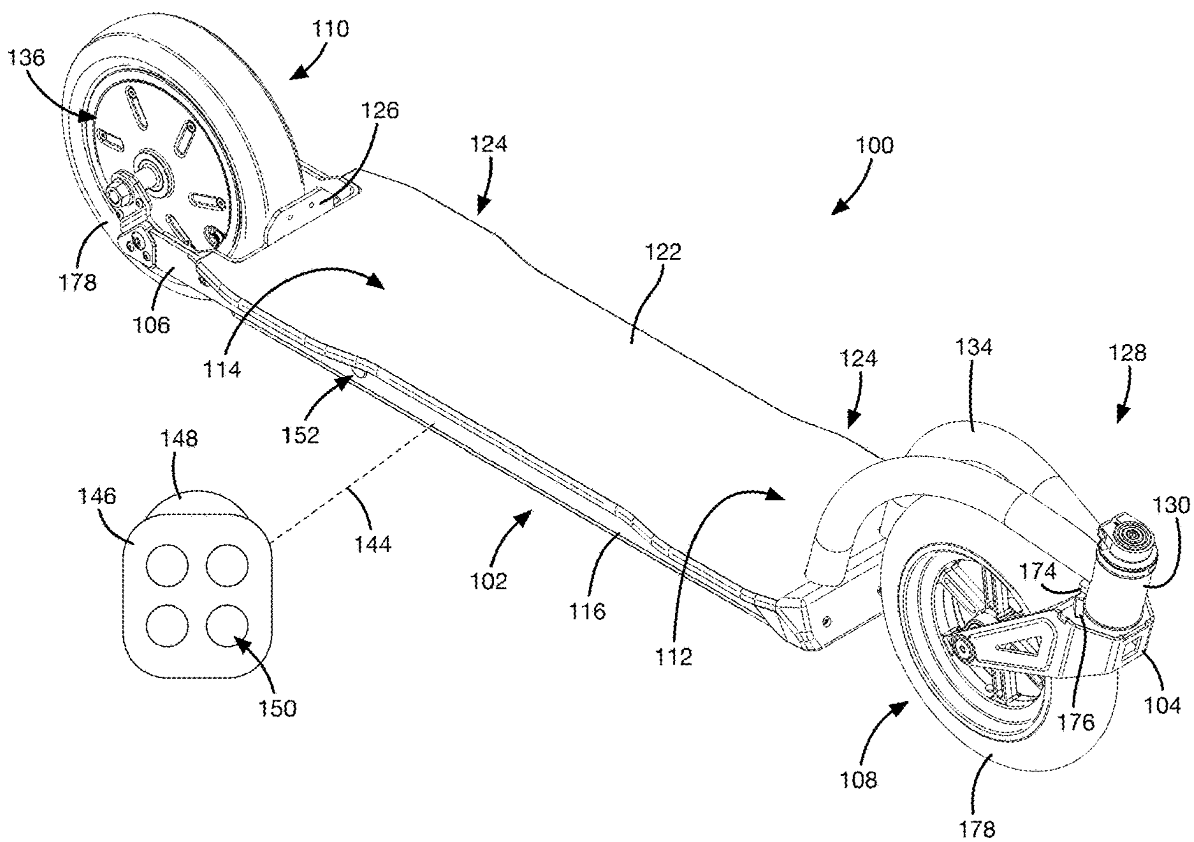
CPC ***B62J 43/10*** (2020.02); ***B62K 21/08*** (2013.01); ***B62K 21/18*** (2013.01); ***B62K 23/02*** (2013.01); *B62K 2202/00* (2013.01)

(57) **ABSTRACT**

A powered two-wheel vehicle includes a chassis extending between a rotatable front fork and a fixed rear fork. A deck is attached to the chassis to enclose a compartment. A front wheel is rotatably attached to the front fork and a rear wheel is rotatably attached to the rear fork via a hub motor. The steering assembly includes a head tube supported above and in front of the deck by a pair of arched frame tubes, a steering tube rotatably connected to the head tube, and a rotational damper connected between the head tube and the steering tube.



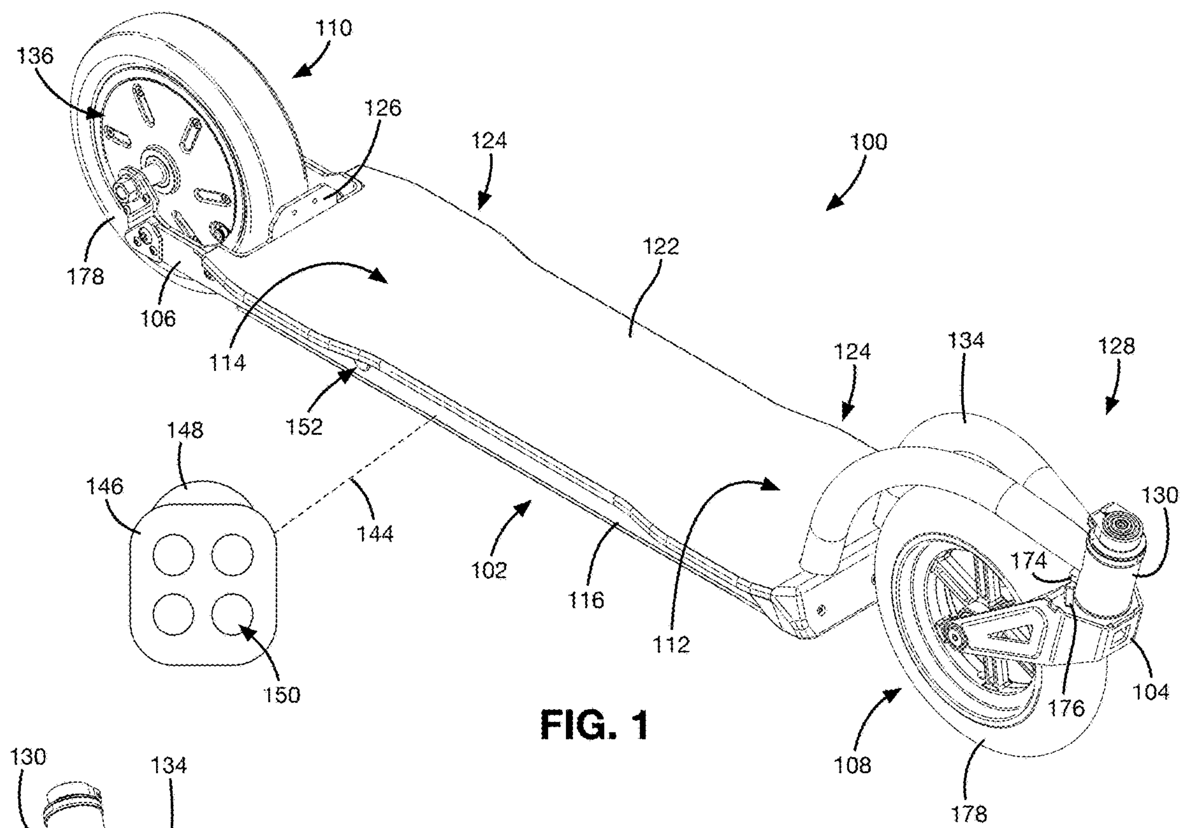

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FIG. 1

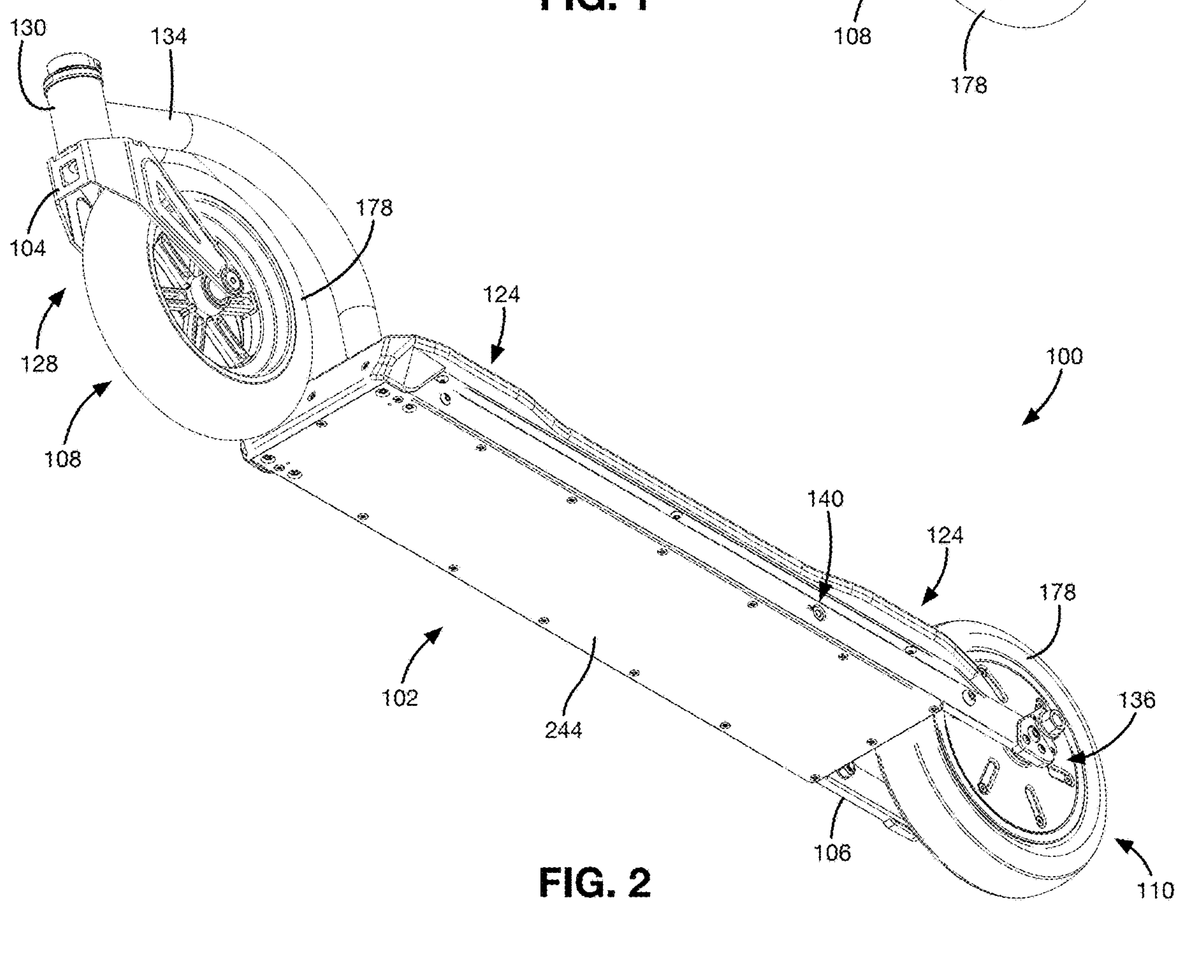
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FIG. 2

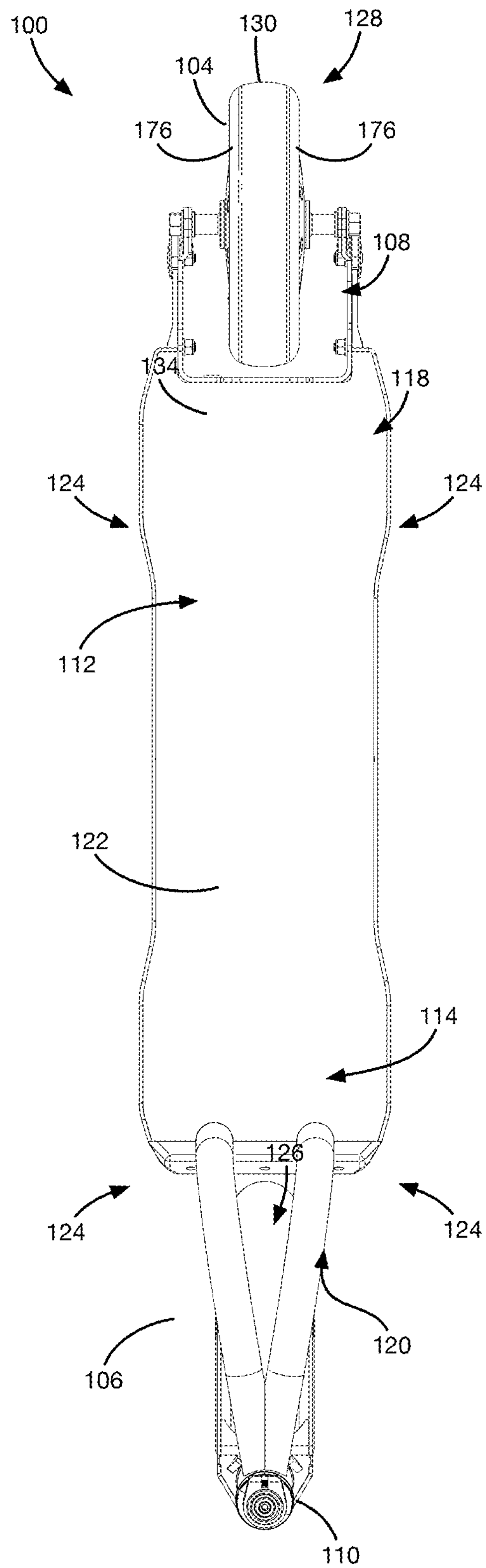
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FIG. 3

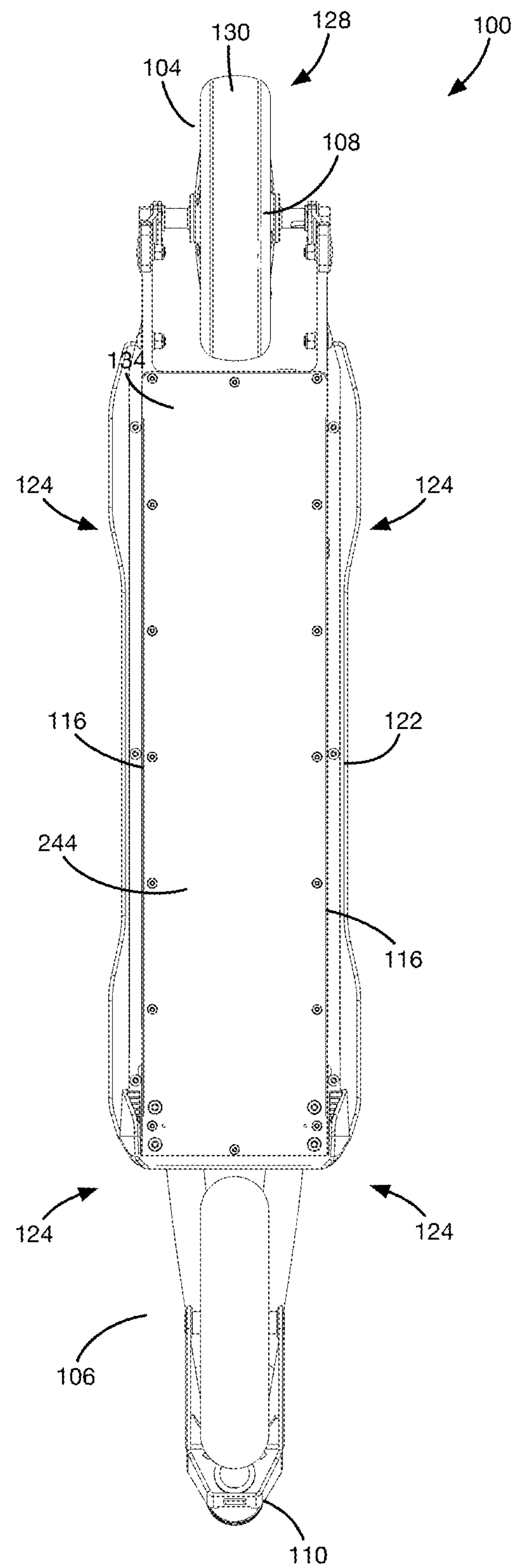
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FIG. 4

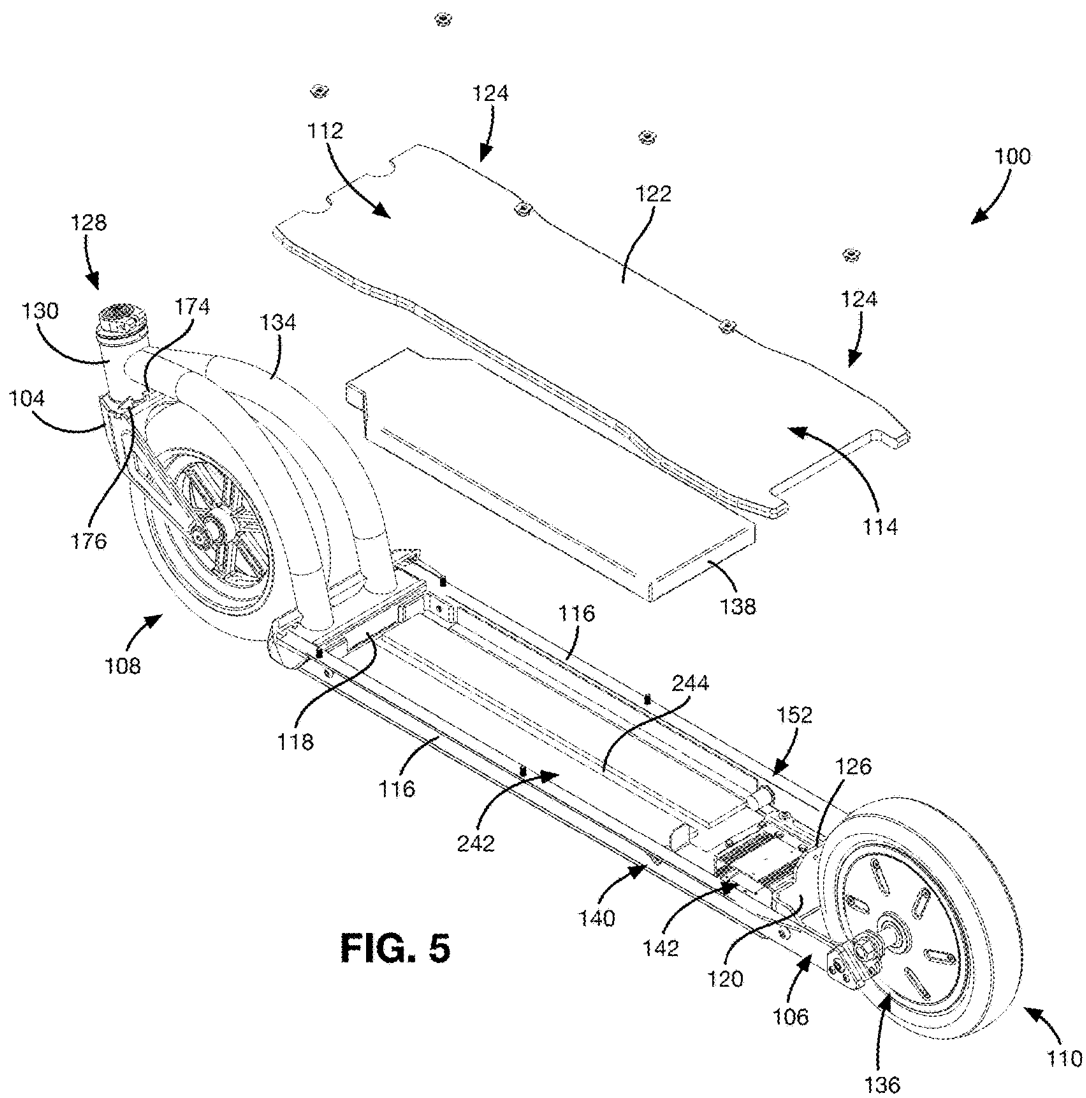
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FIG. 5

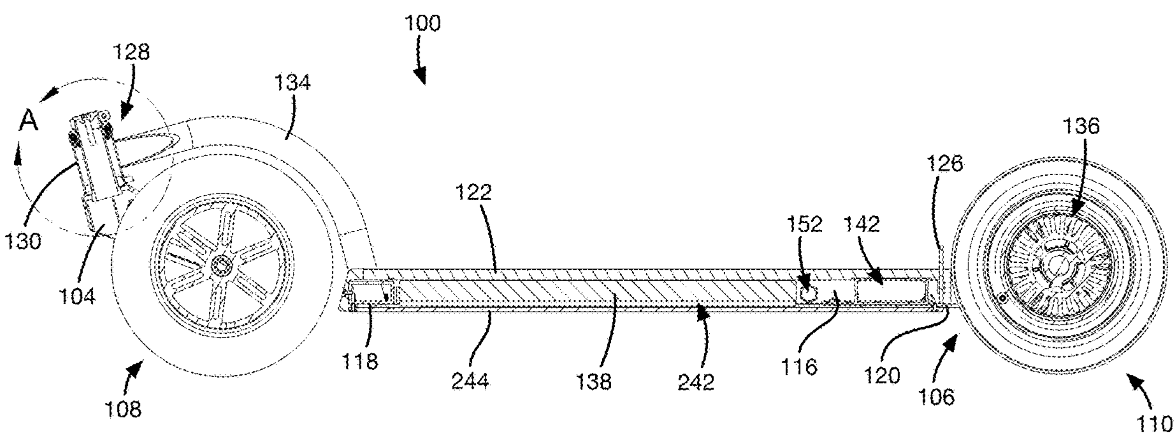
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FIG. 6

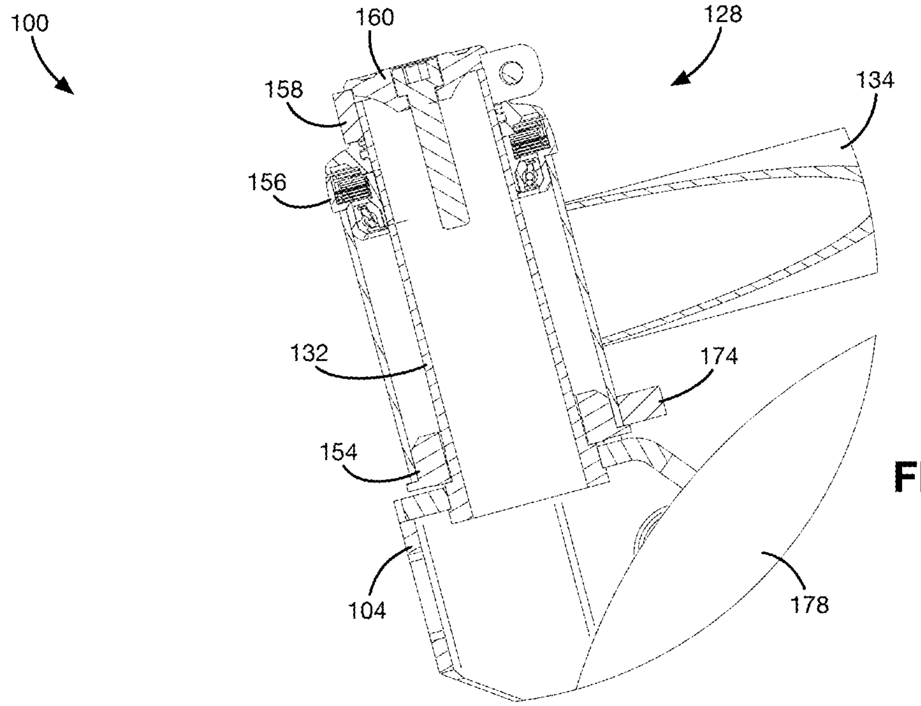
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FIG. 7

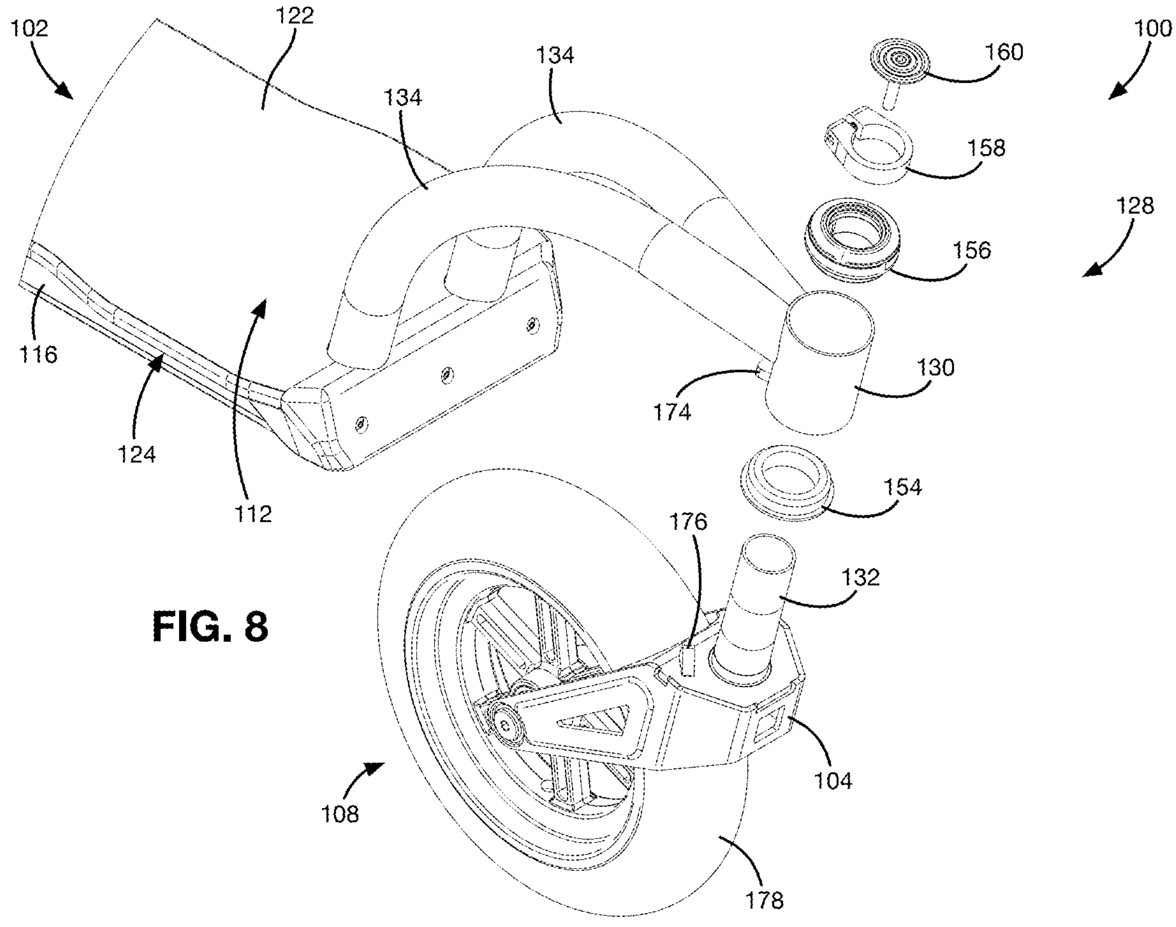
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FIG. 8

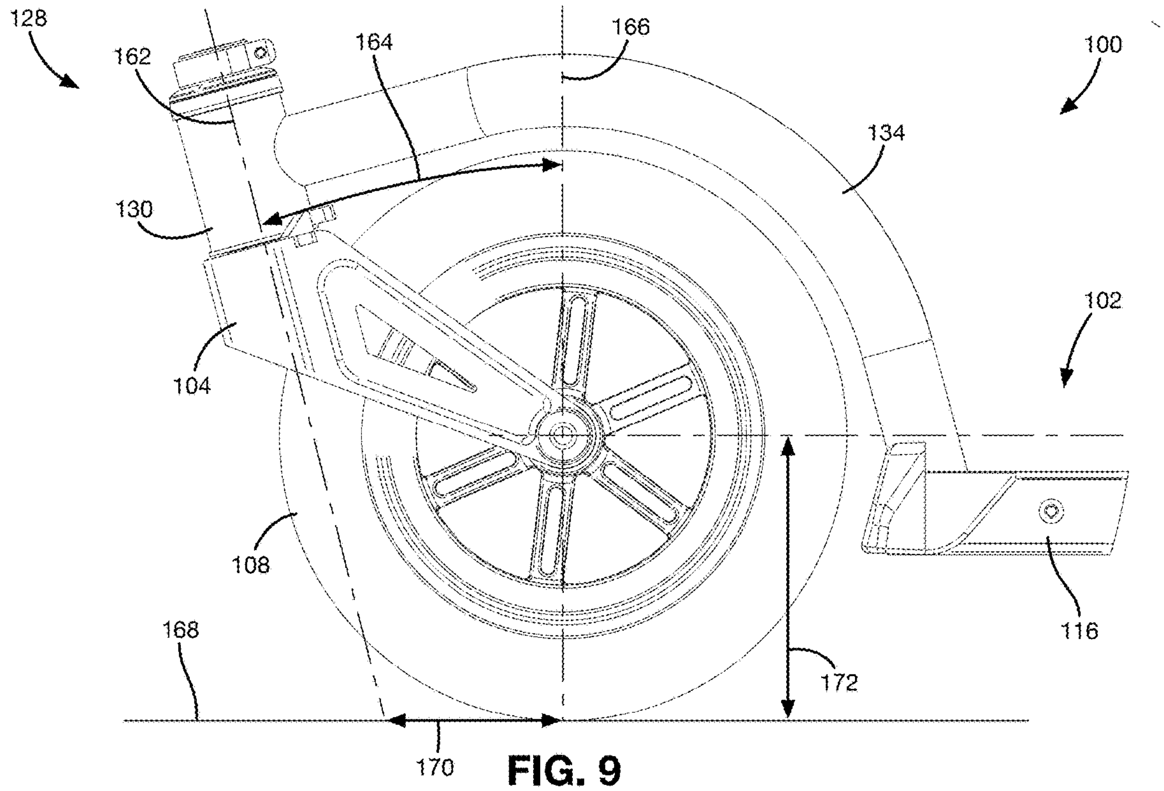
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FIG. 9

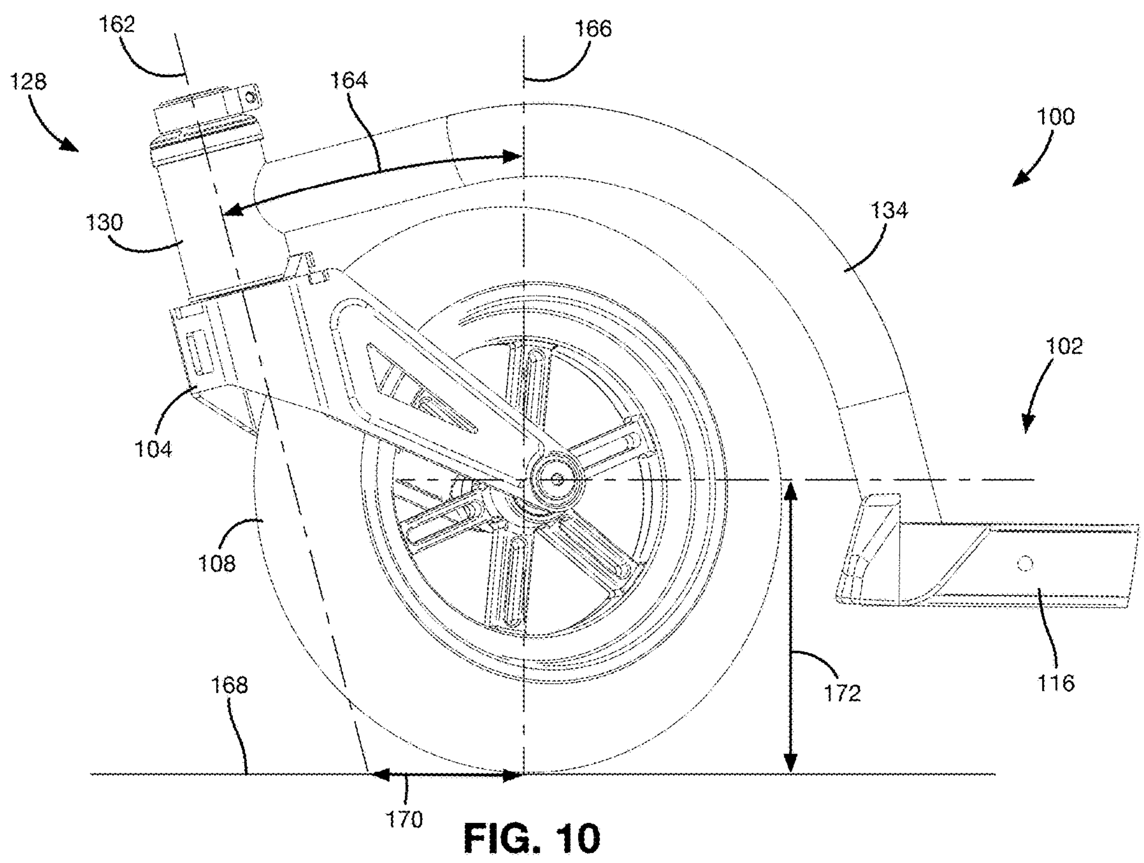
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FIG. 10

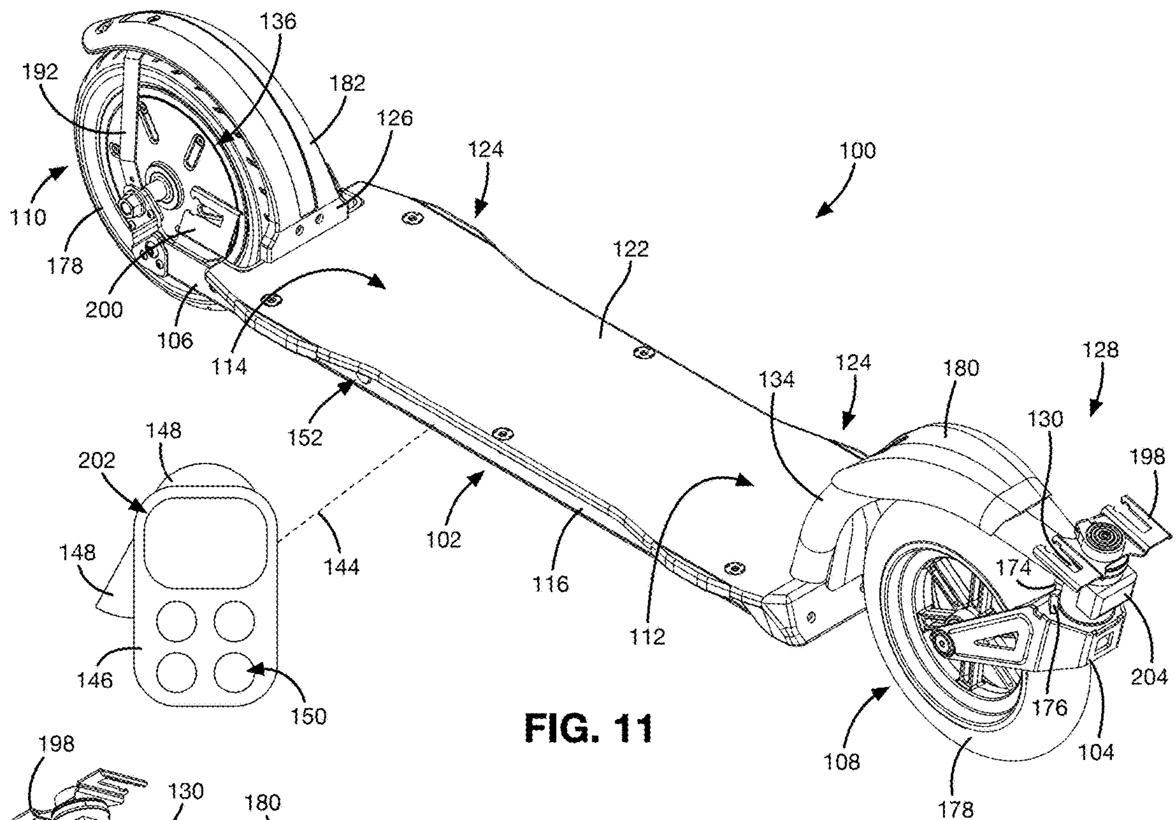
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FIG. 11

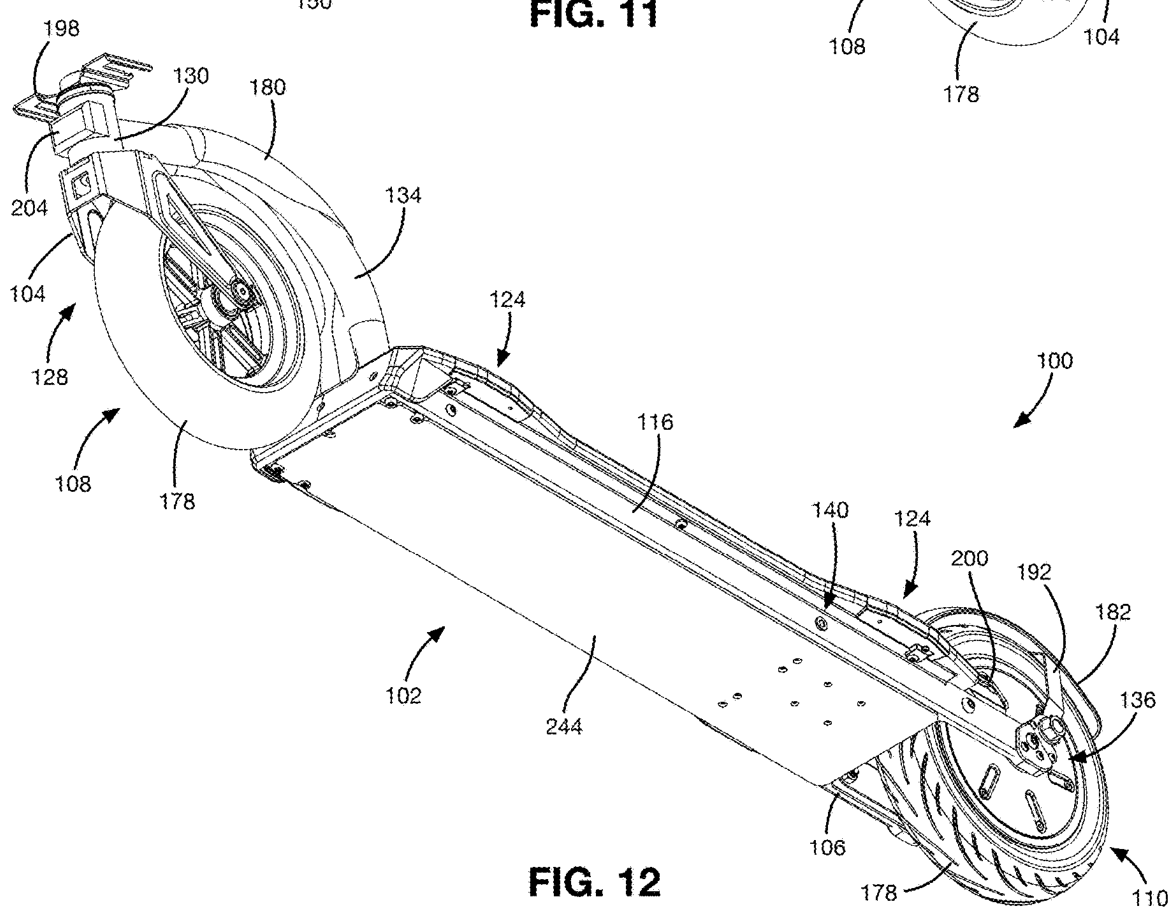
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FIG. 12

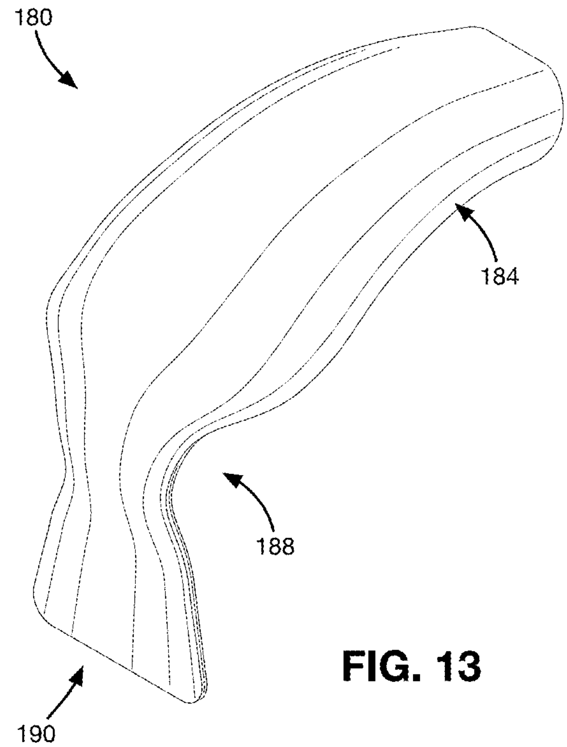

FIG. 13

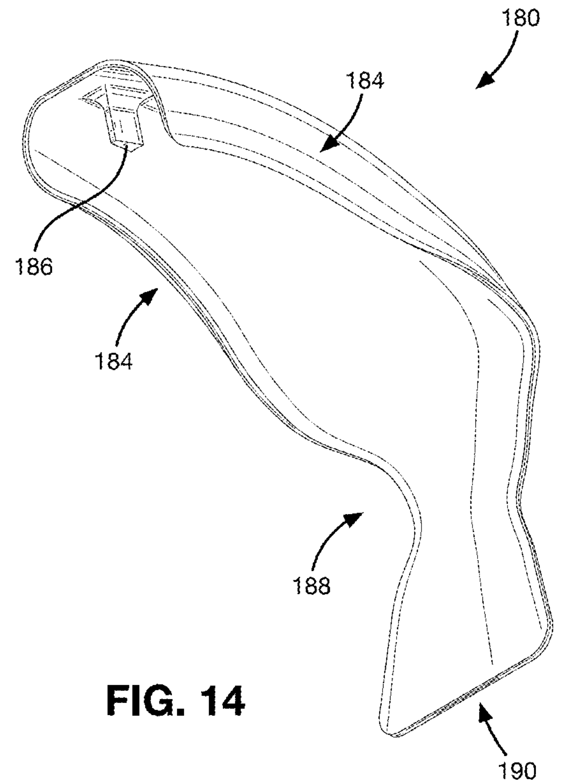

FIG. 14

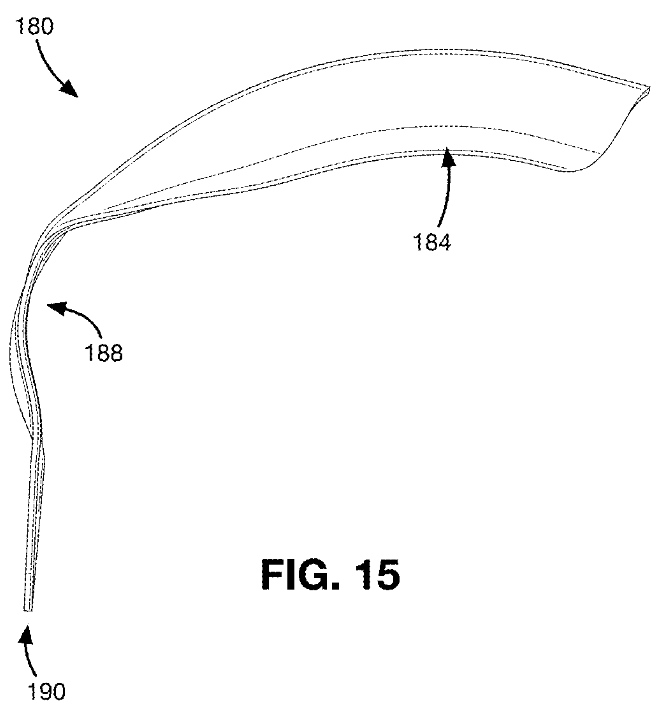

FIG. 15

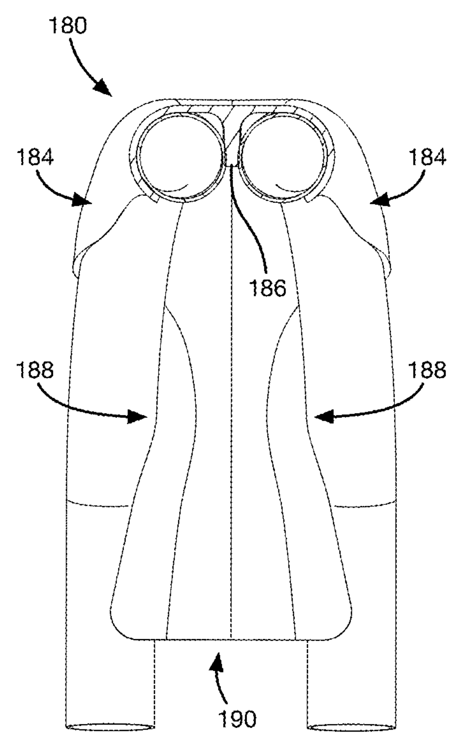

FIG. 16

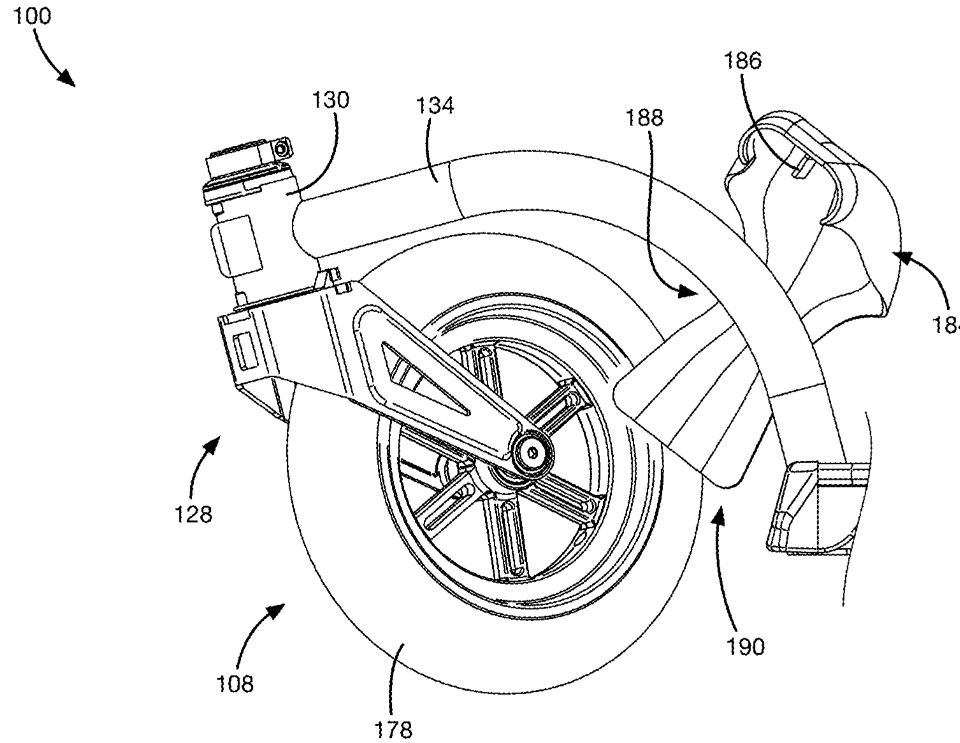
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FIG. 17

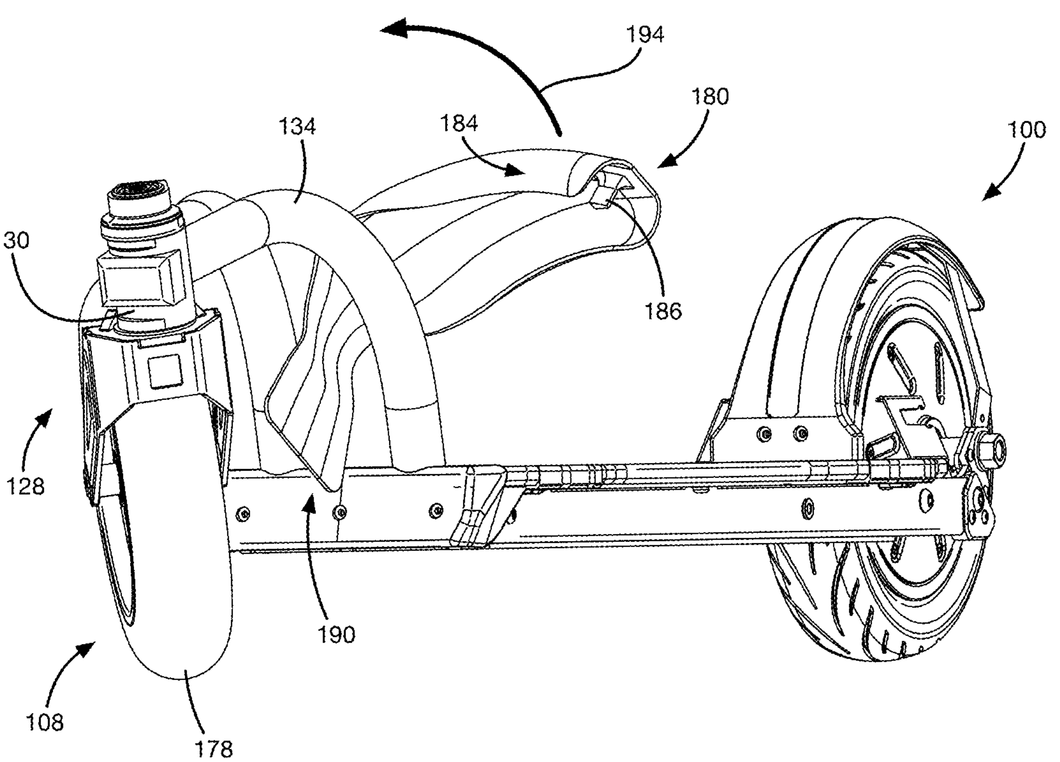
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FIG. 18

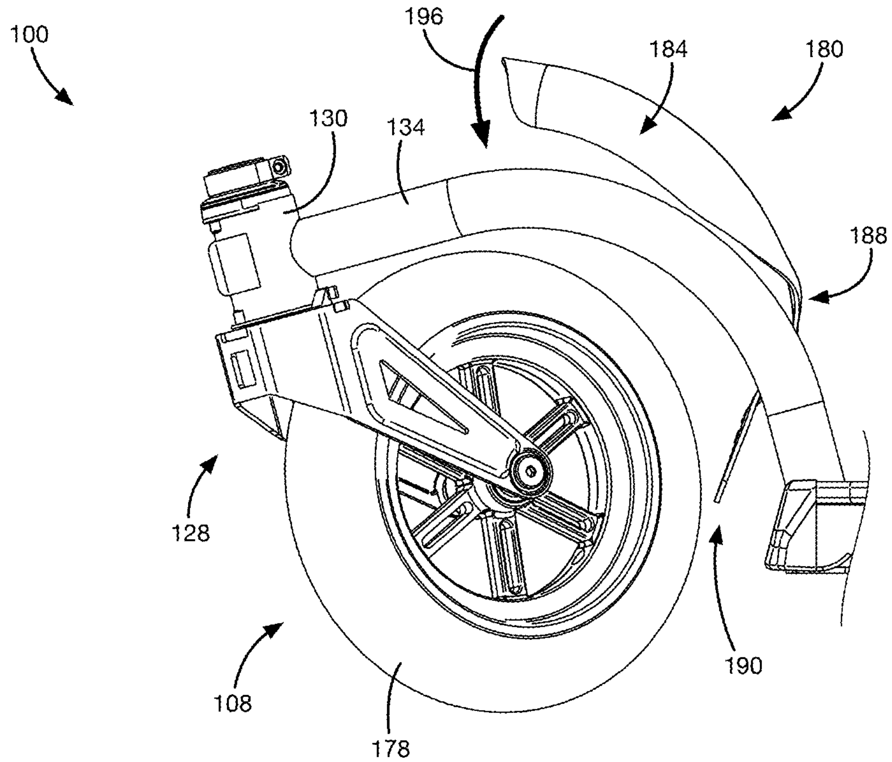
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FIG. 19

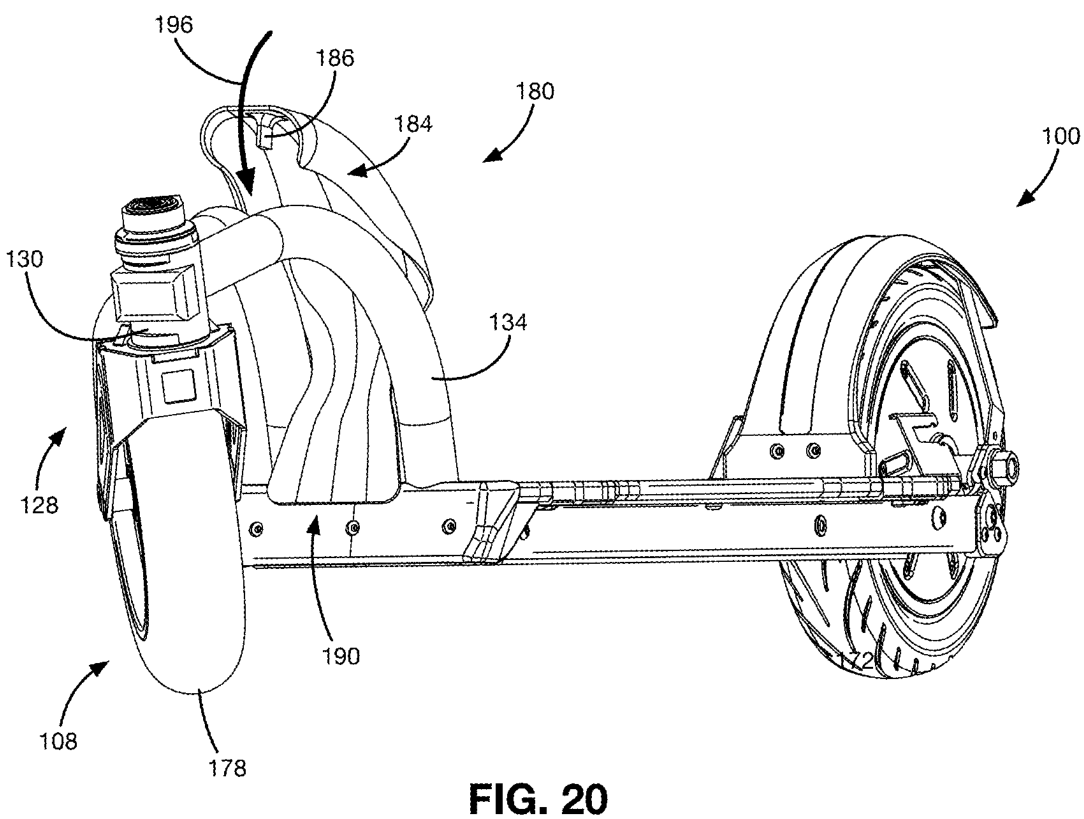
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FIG. 20

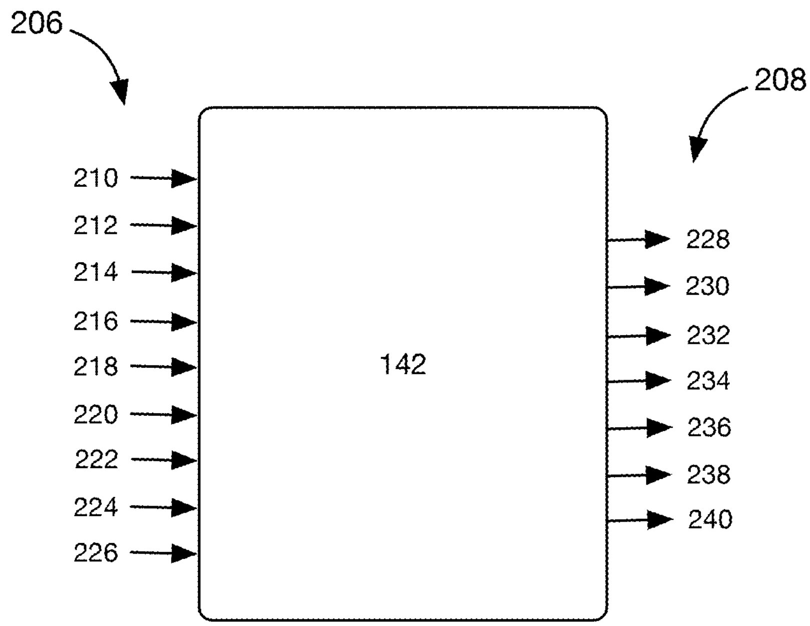
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FIG. 21

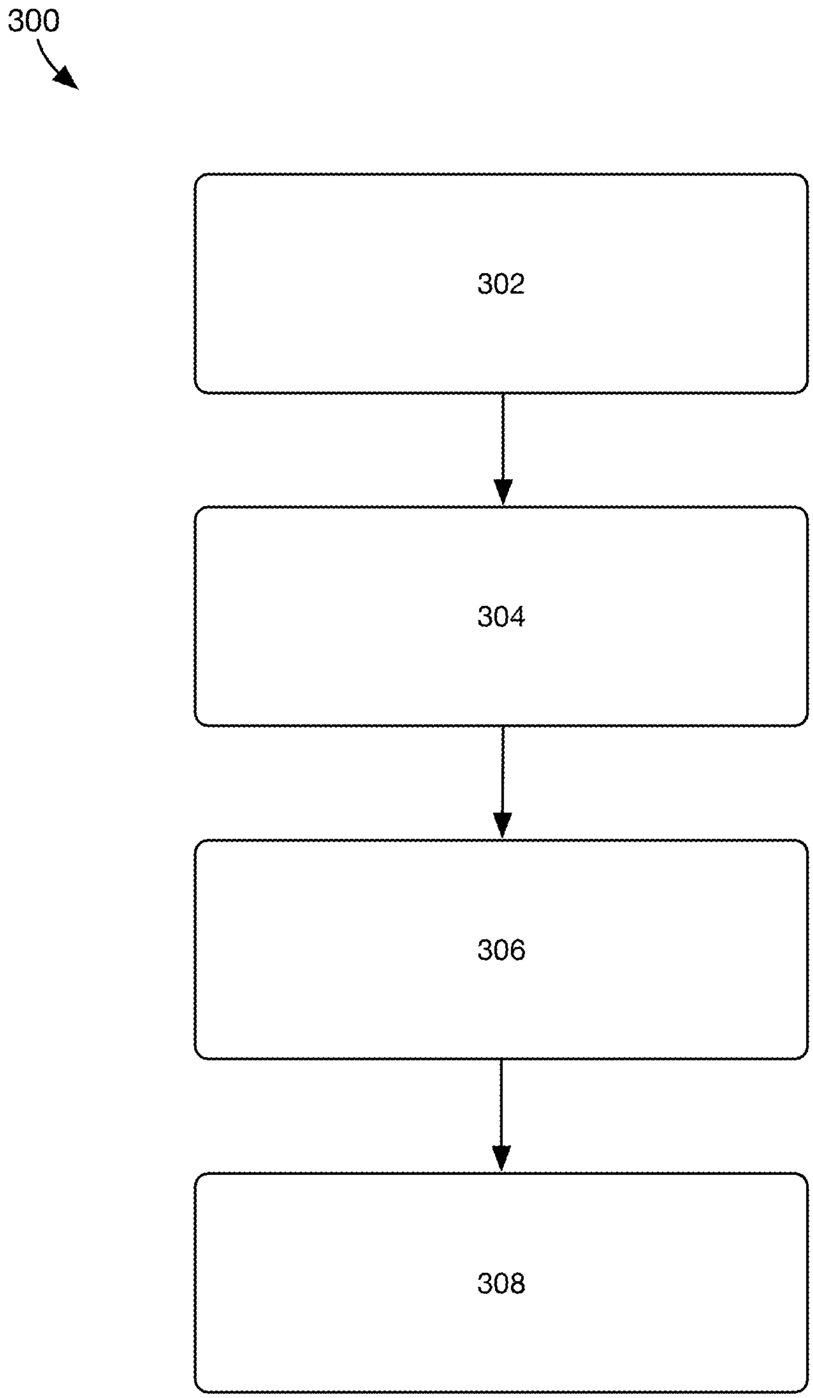

FIG. 22

POWERED TWO-WHEEL VEHICLE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is a continuation of U.S. application Ser. No. 18/405,520, which is related to and claims any benefit of U.S. Provisional Application Ser. No. 63/437,487, filed on Jan. 6, 2023, titled POWERED TWO-WHEEL VEHICLE, the disclosures of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] The present application relates generally to powered mobility devices and in particular to two-wheeled, electrically powered boards.

BACKGROUND

[0003] With the variable price of gasoline along with the current public perception of internal combustion engines and their emissions' negative impact on the environment, there is a growing need in the market for an alternative mode of transportation. In densely populated urban areas, highly congested city streets have led many to use non-motorized modes of transportation that have a smaller footprint, such as a bike or skateboard. Using these smaller, manually powered vehicles allows a rider to avoid the traffic of the main roadway but requires physical exertion unlike a traditional motorized vehicle. These traditional vehicles can be limited in other ways as well, for example, skateboards require a smooth surface to work safely and effectively, can be difficult to learn to maneuver, and can be unstable at higher speeds. As such, there is a need for a motorized mobility device that is compact and lightweight, easy to learn, and stable at higher speeds achieved by a motorized vehicle.

SUMMARY

[0004] Exemplary embodiments of two-wheeled powered vehicles and methods of using the same are disclosed herein.

[0005] An example of a powered two-wheel vehicle includes a chassis, a deck, a front wheel, a rear wheel, and a steering assembly. The chassis extends between a rotatable front fork and a fixed rear fork. The deck is removably attached to the chassis, wherein attaching the deck to the chassis encloses a compartment. The front wheel is rotatably attached to the rotatable front fork. The rear wheel is rotatably attached to the fixed rear fork via a hub motor, wherein the hub motor provides motive power to the rear wheel. The steering assembly includes a head tube, a steering tube, and a rotational damper. The head tube is attached to the chassis and supported above and in front of the deck by a pair of arched frame tubes, wherein the head tube is tilted forward at a rake angle. The steering tube is rotatably connected to the head tube, wherein the steering tube extends into the head tube from the rotatable front fork so that the rotatable front fork trails the head tube when the powered two-wheel vehicle moves in a forward direction and the front wheel is in contact with a ground surface. The rotational damper is connected between the head tube and the steering tube.

[0006] An example of a powered two-wheel vehicle includes a chassis, a deck, a front wheel, a rear wheel, and a steering assembly. The chassis extends between a rotatable front fork and a fixed rear fork. The deck is removably attached to the chassis, wherein attaching the deck to the chassis encloses a compartment. The front wheel is rotatably attached to the rotatable front fork. The rear wheel is rotatably attached to the fixed rear fork via a hub motor, wherein the hub motor provides motive power to the rear wheel. The steering assembly includes a head tube, a steering tube, and a rotational damper. The head tube is attached to the chassis and supported above and in front of the deck by a pair of arched frame tubes, wherein the head tube is tilted forward at a rake angle. The steering tube is rotatably connected to the head tube, wherein the steering tube extends into the head tube from the rotatable front fork so that the rotatable front fork trails the head tube when the powered two-wheel vehicle moves in a forward direction and the front wheel is in contact with a ground surface. The rotational damper is connected between the head tube and the steering tube. The powered two-wheel vehicle also includes a battery and a control system disposed in the compartment. The battery provides electrical power to the hub motor and the control system. The powered two-wheel vehicle also includes a user input device for transmitting user input signals to the control system. The user input device includes a plurality of control elements for controlling the operation of the powered two-wheel vehicle.

[0007] An example method of controlling a powered two-wheel vehicle includes steps of: receiving a user input signal from a user input device at a control system, receiving a selected operating mode and vehicle performance data at the control system, generating a motor control signal, and operating a motor based on the motor control signal. The user input signal includes an acceleration instruction or a deceleration instruction. The motor control signal is generated based on the user input signal, the vehicle performance data, and the selected operating mode.

[0008] A further understanding of the nature and advantages of the present invention are set forth in the following description and claims, particularly when considered in conjunction with the accompanying drawings in which like parts bear like reference numerals.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] To further clarify various aspects of embodiments of the present disclosure, a more particular description of the certain embodiments will be made by reference to various aspects of the appended drawings. It is appreciated that these drawings depict only typical embodiments of the present disclosure and are therefore not to be considered limiting of the scope of the disclosure. Moreover, while the figures can be drawn to scale for some embodiments, the figures are not necessarily drawn to scale for all embodiments. Embodiments and other features and advantages of the present disclosure will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:

[0010] FIG. **1** shows a top perspective view of a powered two-wheel vehicle;

[0011] FIG. **2** shows a bottom perspective view thereof;

[0012] FIG. **3** shows a top view thereof;

[0013] FIG. **4** shows a bottom view thereof;

[0014] FIG. **5** shows an exploded view thereof;

[0015] FIG. **6** shows a cross-sectional view thereof, taken along a midplane of the powered two-wheel vehicle;

[0016] FIG. **7** shows an enlarged view of area A of FIG. **6**;

[0017] FIG. **8** shows an exploded view of a steering assembly of the powered two-wheel vehicle of FIG. **1**;

[0018] FIG. **9** shows an enlarged side view of the front wheel of the powered two-wheel vehicle of FIG. **1** with the front wheel in a straight condition;

[0019] FIG. **10** shows an enlarged side view of the front wheel of the powered two-wheel vehicle of FIG. **1** with the front wheel in a turned condition;

[0020] FIG. **11** shows a top perspective view of a powered two-wheel vehicle;

[0021] FIG. **12** shows a bottom perspective thereof;

[0022] FIG. **13** shows a top perspective view of a front fender of the powered two-wheel vehicle;

[0023] FIG. **14** shows a bottom perspective view thereof;

[0024] FIG. **15** shows a side view thereof;

[0025] FIG. **16** shows a cross-sectional view of the front fender assembled to frame tubes of the powered two-wheel vehicle;

[0026] FIG. **17** shows a side view of the front fender inserted between the frame tubes;

[0027] FIG. **18** shows a left front perspective view thereof;

[0028] FIG. **19** shows a side view of the front fender being attached to the frame tubes;

[0029] FIG. **20** shows a left front perspective view thereof;

[0030] FIG. **21** shows a block diagram of a control system for an exemplary powered two-wheel vehicle; and

[0031] FIG. **22** shows a flow diagram that illustrates an exemplary methodology for controlling a powered two-wheel vehicle.

DETAILED DESCRIPTION

[0032] The following description refers to the accompanying drawings, which illustrate specific embodiments of the present disclosure. Other embodiments having different structures and operation do not depart from the scope of the present disclosure.

[0033] In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of one or more aspects. It may be evident, however, that such aspect(s) may be practiced without these specific details. In other instances, well-known structures and devices are shown in block diagram form in order to facilitate describing one or more aspects. Further, it is to be understood that functionality that is described as being carried out by certain system components may be performed by multiple components. Similarly, for instance, a component may be configured to perform functionality that is described as being carried out by multiple components.

[0034] As described herein, when one or more components are described as being connected, joined, affixed, coupled, attached, or otherwise interconnected, such interconnection may be direct as between the components or may be indirect such as through the use of one or more intermediary components. Also as described herein, reference to a "member," "component," or "portion" shall not be limited to a single structural member, component, or element but can include an assembly of components, members, or elements. Additionally, as used herein, the term "exemplary" is intended to mean serving as an illustration or example of something and is not intended to indicate a preference.

[0035] Moreover, the term "or" is intended to mean an inclusive "or" rather than an exclusive "or." That is, unless specified otherwise, or clear from the context, the phrase "X employs A or B" is intended to mean any of the natural inclusive permutations. That is, the phrase "X employs A or B" is satisfied by any of the following instances: X employs A; X employs B; and X employs both A and B. In addition, the articles "a" and "an" as used in this application and the appended claims should generally be construed to mean "one or more" unless specified otherwise or clear from the context to be directed to a singular form.

[0036] Numerical values or ranges stated herein are understood to encompass values at or near the stated value and/or above or below the stated range. For this application, the stated value can encompass plus or minus 5% of the value and the stated range can encompass plus or minus 5% of the extent of the range. In addition, the stated value or range can include a margin of error for the value or range typical in the art for the property being measured. The stated value or range can also encompass those values and ranges that would be considered equivalent to the stated value or range by one of ordinary skill in the art. As an example, a voltage expressed as a range of 10 volts to 100 volts is understood to include durations above and below the ends of the range by 5% of the extent of the 90-volt range—e.g., 6.5 volts to 104.5 volts. As another example, the size of an object expressed as a value of 2 inches includes values above and below 2 inches that are within the margin of error of a tool typically used to measure objects of that size. As yet another example, an angle between two components expressed as a value of 20 degrees includes values above and below 20 degrees that would be considered equivalent by one of ordinary skill in the art.

[0037] Further, as used herein, the terms "component" and "system" are intended to encompass computer-readable data storage that is configured with computer-executable instructions that cause certain functionality to be performed when executed by a processor. The computer-executable instructions may include a routine, a function, or the like. It is also to be understood that a component or system may be localized on a single device or distributed across several devices.

[0038] Moreover, the acts described herein may be computer-executable instructions that can be implemented by one or more processors and/or stored on a computer-readable medium or media. The computer-executable instructions can include a routine, a sub-routine, programs, a thread of execution, and/or the like. Still further, results of acts of the methodologies can be stored in a computer-readable medium, displayed on a display device, and/or the like.

[0039] Since their introduction in the early 1960s, skateboards have garnered a strong following and culture unique from other forms of personal transportation. While a typical skateboard that has four wheels on two axels attached to the bottom of the board or deck can be used for transportation, the maneuverability of the four-wheel layout is limited in that the four wheels discourage sharp turns. Riders get around this limitation by performing various tricks that lift one or more wheels off of the ground. Learning how to perform these maneuvers requires an amount of practice and possibility of injury that dissuades the average pedestrian from using a skateboard for a daily commute.

[0040] While skateboards and other small, manually powered vehicles such as scooters, skates, and bicycles allow a pedestrian to avoid the traffic of the main roadway, the physical exertion required by these devices can dissuade one from using such a device to commute to work or to travel between errands. Reducing the physical exertion required to ride, for example a skateboard, can be done by the addition of a motor to drive one or more of the wheels of the skateboard. The small, hard wheels and shorter wheelbase of the skateboard, however, results in reduced stability as the rider increases in speed and limits the use of the skateboard to paved surfaces. The powered two-wheel vehicle described herein provides improved maneuverability—riding more like a surfboard or snowboard—and better stability at higher speeds than a typical skateboard. An example of a powered two-wheel vehicle described herein has a tighter turning radius than a typical skateboard at low speeds, is stable at higher speeds, and can traverse unpaved paths, lawns, and other rough terrain. Importantly, the stability of the powered two-wheel vehicles described herein helps new riders quickly learn how to control the vehicle, thereby facilitating increased adoption in the marketplace. An optional training post can also be added to the powered two-wheel vehicle described herein to provide a new rider with an added sense of security when first stepping onto the powered two-wheel vehicle.

[0041] Referring now to FIGS. **1-10**, an example of a powered two-wheel vehicle **100** is shown. The powered two-wheel vehicle **100** includes a chassis **102** that extends between a rotatable front fork **104** and a fixed rear fork **106**. A front wheel **108** is rotatably attached to the rotatable front fork **104** and a rear wheel **110** is rotatably attached to the fixed rear fork **106**. The front wheel **108** and the rear wheel **110** are arranged inline and each have a diameter ranging from 5 inches to 17 inches, or from 6 inches to 15 inches, or from 7 inches to 13 inches, or from 8 inches to 11 inches, or 9.5 inches. In some powered two-wheel vehicles, the front and rear wheels **108**, **110** have the same diameter such that a rotational center of each wheel **108**, **110** is the same height above the ground. In some powered two-wheel vehicles, the front wheel **108** and the rear wheel **110** have different diameters—i.e., the front wheel **108** can have a larger diameter than the rear wheel, and vice versa.

[0042] The inline arrangement of the wheels **108**, **110** allows the rider to turn with a tighter turning radius than a vehicle with four wheels, providing enhanced maneuverability for the powered two-wheel vehicle **100** that is helpful in urban and other crowded environments. The powered two-wheel vehicle **100** is steered by the rider leaning forwards and backwards to shift their weight to either side of a centerline of the powered two-wheel vehicle **100**. As the rider shifts their weight to one side or the other of the centerline, the front wheel **108** is pivoted in the direction that the rider leans to facilitate turning. In this way, the powered two-wheel vehicle **100** can be steered only by the rider shifting their weight. Applicant notes that at lower speeds, when the rider leans to one side the front fork **104** and front wheel **108** turn to facilitate steering of the powered two-wheel vehicle **100**. As speed increases, gyroscopic forces of the rotation of front wheel **108** dominate such that the lean of the rider causes the powered two-wheel vehicle **100** to lean to one side so that the point where the ground contacts the front and rear wheels **108**, **110** is moved away from the center and towards the side of the front and rear wheels **108**, **110**. Thus, the rider can “carve” when steering the powered two-wheel skateboard **100** like a snowboard or surfboard.

[0043] The front wheel **108** and the rear wheel **110** each include tires **178** that have a rounded profile to facilitate the carving turning behavior. The rounded profile of the tire **178** provides a sufficient contact area with the ground while the wheel **108**, **110** leans during a turn so that the tire **178** does not slip laterally. The tires **178** can be formed from any suitable material and can be solid, inflated, or take on any other suitable form. The exemplary powered two-wheel vehicle **100** described herein includes tires **178** that are inflated to act as a suspension absorbs impacts with the ground surface to provide a smoother ride for the rider.

[0044] The chassis **102** extending between the rotatable front fork **104** and a fixed rear fork **106** provides support for a first foothold **112** and a second foothold **114** on which the rider can stand. The chassis **102** also provides a rigid connection between the front wheel **108** and the rear wheel **110** and can be formed from a single component or from many components. For example, the chassis **102** shown in FIG. **5** includes a pair of side beams **116** extending between a front bulkhead **118** and a rear bulkhead **120** and a bottom plate **244** that extends between the two side beams **166**. The bottom plate **244** can be attached to the side beams **116** and the front and rear bulkheads **118**, **120** to enclose the bottom of the chassis **102** and provide additional stiffness to the powered two-wheel vehicle **100**. The side beams **116**, front and rear bulkheads **118**, **120**, and bottom plate **244** can be rigidly connected via any suitable means, such as, for example, welding, fasteners, brackets, and the like. The side beams or rails **116** can be formed from tubes, extruded shapes, or sheet metal bent into an appropriate shape, such as a C-shape and can extend beyond the front and rear bulkheads **118**, **120** and be integrally formed with other components of the powered two-wheel vehicle **100**, such as, for example, the fixed rear fork **106** or a pair of frame tubes **134** that support a head tube **130** of a steering assembly **128**.

[0045] The chassis **102** includes a compartment **242** for carrying and protecting various components of the powered two-wheel vehicle **100**. The compartment **242** can be formed between side beams or rails **116**, the bulkheads **118**, **120**, and the bottom plate **224** and can be covered and enclosed by a deck **122**. The compartment **242** can be accessed via removal of the bottom plate **224** and/or the deck **122**. In an example of a powered two-wheel vehicle **100**, the compartment **242** is a water-tight compartment that includes one or more seals or gaskets to prohibit the ingress of water, dirt, and debris from entering the compartment **242** and potentially damaging any sensitive components stored therein.

[0046] In a chassis **102** formed from a single piece of material, the side beams or rails **116**, the bulkheads **118**, **120**, and the bottom plate **244** can be formed as portions of the single piece of material and the deck **122** can be removably attached to the chassis **102**. Alternatively, the deck **122** can be integrally formed as part of the chassis **102** and the bottom plate **244** can be removably attached to the chassis **102**. Integrally forming the elements of the chassis **102** from a single piece of material can be accomplished by any suitable means, such as, for example, by machining, forging, injection molding, stamping, and the like. In examples of powered two-wheel vehicles **100** with a single-piece chassis **102**, the compartment **242** can be formed without seams or joints that could leak.

[0047] The bottom plate **244** can also act as a skid plate that facilitates sliding over curbs, rails, or other obstacles in the environment during the operation of the powered two-wheel vehicle **100**. The bottom plate **244** can include a wear coating or a replaceable wearable component so that sliding across the bottom plate **244** does not damage the material of the bottom plate **244** and chassis **102**. In an example of a powered two-wheel vehicle **100**, the bottom plate **244** is made from aluminum includes a removable skid plate (not shown) that is attached to the bottom plate **244** and is made from, for example, ultra-high molecular weight plastic (UHMW) that is more resistant to wear than the aluminum of the bottom plate **244**. A removable skid plate allows the skid plate to be easily replaced when it is too worn and can no longer protect the bottom plate **244**.

[0048] The first foothold **112** and second foothold **114** provide locations for the rider to place each of their feet and can have a high friction surface for providing additional friction between the first and second footholds **112**, **114** and the feet of the rider. The first foothold **112** and the second foothold **114** can take on a wide variety of forms, such as, for example, foot bindings like those typically seen on snowboards, clips for receiving specially configured boots or shoes, or a single deck, such as the deck **122**. The deck **122** is attached to the chassis **102**, has a length of 26 inches to 38 inches, and is covered with a high friction material, such as grip tape. In an example of a powered two-wheel vehicle **100**, the deck **122** is attached to both of the side beams **116** and the bottom plate **244** is attached to both of the side beams **116** and the bulkheads **118**, **120** so that the components of the chassis **102** form a torsion box arrangement that provides a stable and stiff platform for the powered two-wheel vehicle **100** that resists twisting that could be detrimental to the ride quality and precision control offered by the powered two-wheel vehicle **100**.

[0049] The deck **122** can include widened portions **124** that provide additional support for the feet of the rider and can act as handles when the rider is in a crouching position. While the deck **122** is shown as a single, continuous piece of material, the deck **122** can include various openings or be formed from multiple pieces spaced apart around an opening to accommodate other components of the powered two-wheel vehicle **100**, such as, for example, a motor, a power source (e.g., a battery), a handle, or the like. The deck **122** can optionally include a seal (not shown) that prohibits ingress of water or other contaminants into the interior spaces of the powered two-wheel vehicle **100** that can include electronic components. A guard **126** extends from the rear bulkhead **120** at a rear end of the deck **122** to prohibit contact between the rider's foot and the rear wheel **110**.

[0050] The height of the deck **122** below the height of the centers of the wheels **108**, **110**—i.e., the wheel radius—can be 0.125 inches to 5 inches. The top of the deck **122** can also be even with the center of the wheels **108**, **110**, that is, the height of the deck **122** above the ground can be equal to the radius of the front and rear wheels **108**, **110**. As the deck **122** is lowered relative to the centerline of the wheels **108**, **110** the stability of the rider on the powered two-wheel vehicle **100** is increased as the weight of the rider is applied to a surface below a pivot point of the powered two-wheel vehicle **100** which provides inherent stability against leaning as the center of mass of the rider is moved closer to the rotational axis of the wheels **108**, **110**. However, reducing a ground clearance measured between the bottom of the powered two-wheel vehicle **100** and the ground can limit the size of bumps or rough terrain the powered two-wheel vehicle **100** can traverse without getting stuck or damaged. Thus, a balance must be struck between the height of the deck **122** and the ground clearance below the chassis **102**.

[0051] The rotatable front fork **104** is attached to the chassis **102** via a steering assembly **128** that includes the head tube **130** fixedly attached to the chassis **102**. The steering post or tube **132** extends from the rotatable front fork **104** and into the head tube **130** to facilitate rotation of the rotatable front fork **104** relative to the chassis **102**. As is described in greater detail below, the steering assembly **128** provides passive stability to the powered two-wheel vehicle **100** while traveling across all kinds of terrain. The head tube **130** is supported above and ahead of the front wheel **108** by two arched frame tubes **134** extending from the front bulkhead **118** of the chassis **102**. The frame tubes **134** have an arched shape to provide clearance for the front wheel **108** to rotate from side-to-side as the powered two-wheel vehicle **100** is turned by the rider. An optional mud guard (not shown) can be attached to the frame tubes **134** at the front of the powered two-wheel vehicle **100** and the guard **126** or rear fork **106** at the rear of the powered two-wheel vehicle **100**. In another configuration, an optional training post (not shown) can extend vertically from the front bulkhead **118** between the frame tubes **134** to a handle (not shown) that the rider can grip for added stability and steering control when learning to ride the powered two-wheel vehicle **100**. The handle of the training post can have an adjustable height to accommodate riders of different heights.

[0052] A motor **136** provides motive power to one or both of the front wheel **108** and the rear wheel **110**. The motor **136** of the powered two-wheel vehicle **100** is an electric hub motor that is mounted inside the rear wheel **110** and is attached to the rear fork **106**. Alternatively, the motor **136** can be mounted to the chassis **102** and be connected to one or both of the wheels **108**, **110** via a transmission. The motor **136** can be an electric motor, as described herein, or can be an internal combustion engine running on fuel supplied from a fuel tank, or any other suitable means of supplying motive power to the wheels **108**, **110**. Control of the motor **136** can be accomplished by applying and releasing tension on a throttle cable for an internal combustion engine or can involve adjustments to the current and voltage provided to the motor **136** for an electrical motor. When the powered two-wheel vehicle **100** is unpowered, the rear wheel **110** is allowed to rotate freely so that the powered two-wheel vehicle **100** can be rolled along the ground without resistance. To transport the powered two-wheel vehicle **100**, the user can grip one of the frame tubes **134** to lift the front wheel **108** off of the ground so that the powered two-wheel vehicle **100** rolls on the rear wheel **110** and the user only has to lift half of the weight of the entire vehicle.

[0053] For the powered two-wheel vehicle **100** described herein, electrical power is provided to the motor **136** from a battery **138**. The battery **138** can be contained within the compartment **242** that is enclosed by the bottom plate **244**, the side beams **116**, and the bulkheads **118**, **120** of the chassis **102**, and the deck **122**. The compartment **242** can be formed when the deck **122** or the bottom plate **244** are attached to the chassis **102**. In some embodiments, the bottom plate **244** and/or the deck **122** are removable to provide access to the compartment **242**. To prohibit water and other foreign matter from entering the compartment **242** and damaging components arranged therein, the compartment **242** can be a water-tight compartment **242** that includes one or more seals or gaskets. The battery **138** can be any suitable battery that provides sufficient power to the motor **136**—e.g., a continuous wattage ranging from 400 watts to 3000 watts—and can have a working voltage ranging from 24 volts to 90 volts. The battery **138** is rechargeable via a charging port **140** provided in one of the side beams **116** of the chassis **102**.

[0054] A control system **142** of the powered two-wheel vehicle **100** receives data from a wide variety of input signals **206** and generates output signals **208** based on those inputs and based on the programming of the control system **142**. The control system **142** includes one or more processors and memory for storing data and programming instructions. The control system **142** can be arranged in the compartment **142** along with the battery **138**. The data stored in the control system **142** can include data related to the current state of the powered two-wheel vehicle **100**, historical performance data, computer-executable instructions for controlling the powered two-wheel vehicle **100**, and the like. The instructions may be, for instance, instructions for implementing functionality described as being carried out by one or more components discussed herein or instructions for implementing one or more of the methods described herein. The operation of the control system **142** to receive a wide variety of input signals and to generate a wide variety of output signals is described in greater detail and with respect to FIG. **21**, below.

[0055] One example of an input received by the control system **142** is a user input signal **144** from a user input device **146**. The control system **142** processes the user input signal **144** and can then output a motor control signal to the motor **136** to cause the powered two-wheel vehicle **100** to, for example, accelerate or decelerate. The user input signal **144** can be transmitted to the control system **142** of the powered two-wheel vehicle **100** wirelessly, as is shown in FIG. **1**, or via a cable or wire connecting the user input device **146** to the control system **142**. The user input device **146** can optionally be secured to the training post described above. The user input device **146** can include a wide variety of control elements such as, for example, buttons, knobs, wheels, triggers, sliders, or the like. The user input device **146** can also include feedback elements, such as, for example, a display **202** (FIG. **11**), a speaker, a vibration motor, and the like for providing visual, audible, and tactile feedback to the rider based on, for example, user input and the conditions of the powered two-wheel vehicle **100**.

[0056] The user input device **146** shown in FIGS. **1** and **11** includes a control wheel **148** that the rider can rotate in a first direction to accelerate to increase speed and a second direction to decelerate or brake. Braking can be accomplished via a mechanical brake or via regenerative braking that recharges the battery **138** from the motor **136** being rotated by the rear wheel **110**. One or more buttons **150** can also be included on the user input device **146** that allow the rider to change an operating mode of the powered two-wheel vehicle **100** (e.g., for different terrain or power setting—limiting the maximum current supplied to the motor) or to switch the orientation or handedness of the user input device **146** depending on whether the rider wants to use their left hand or right hand to control the powered two-wheel vehicle **100** via the user input device **146**. The user input device **146** can also include sensors such as, for example, an accelerometer, gyroscope, or the like that measures the acceleration or tilt angle of the rider to provide additional input to the control system **142**. A power button **152** can be provided on the powered two-wheel vehicle **100** to allow the rider to turn the powered two-wheel vehicle **100** on and off. Alternatively, the user input device **146** can be used to activate the powered two-wheel vehicle **100**.

[0057] Referring now to FIGS. **7-10**, the steering assembly **128** is shown in greater detail. As was noted above, the steering assembly **128** includes the head tube **130** that is supported by two frame tubes **134** that extend from the front bulkhead **118** of the chassis **102**. A bushing or bearing **154** is arranged at a bottom end of the head tube **130** for receiving the steering tube **132** extending from the front fork **104**. The weight of the powered two-wheel vehicle **100** and rider is transferred to the front wheel **108** via the steering assembly **128** and bushing or bearing **154** that is made from a lubricious material or includes ball bearings to reduce friction between the steering tube **132** and the head tube **130**. The steering tube **132** extends through a rotational damper **156** arranged in the head tube **130** and is secured within the head tube **130** by a clamp **158** and a cap **160** that are secured to an upper end of the steering tube **132**. The cap **160** can be attached to a nut (not shown) inside of the steering tube **132**.

[0058] The head tube **130** is arranged having an axis **162** that is arranged at a rake angle **164** relative to a vertical line **166** perpendicular to the side beams or rails **116** of the chassis **102**. The axis **162** points downward and rearward from the top of the head tube **130** such that the rake angle **164** is considered a negative rake angle. The magnitude of the rake angle **164** of the head tube **130** can be in a range of 0 degrees to 25 degrees, or from 5 degrees to 20 degrees, or from 10 degrees to 15 degrees. The magnitude of the rake angle **164** of the illustrated head tube **130** is 15 degrees.

[0059] The negative rake angle **164** of the head tube **130** and the offset of the front fork **104** causes the front wheel **108** to contact a ground plane **168** behind the axis **162** of the head tube **130** so that the front wheel **108** is said to be trailing the head tube **130**. A trailing distance **170** is defined between the point where the front wheel **108** contacts the ground plane **168** and the point where the axis **162** intersects the ground plane **168**. The trailing distance **170** of the powered two-wheel vehicle **100** is in a range of 2 inches to 7 inches. Increasing the trailing distance **170**—e.g., by lengthening the front fork **104**—reduces the effect known as "caster flutter" that can occur as the front wheel **108** is driven across the ground. The front fork **104** can be lengthened by increasing the distance between the steering tube **132** and the axle of the front wheel **108**, for example, by extending the length of the arms of the front fork **104**. The trailing distance **170** can also be impacted by the magnitude of the rake angle and the diameter of the wheel.

[0060] Referring now to FIGS. **9** and **10**, the front wheel **108** is shown in a straight condition (FIG. **9**) and a turned condition (FIG. **10**). An axle height **172** of the front wheel **108** is measured between the ground plane **168** and the axis of rotation of the front wheel **108**. Because the head tube **130** of the steering assembly **128** has a negative rake angle **164**, rotation of the front fork **104** and front wheel **108** causes the axle height **172** of the front wheel **108** to increase as can be seen in FIG. **10**. The weight of the rider and the powered two-wheel vehicle **100** oppose the increase in axle height **172** and work together to encourage the front fork **104** and front wheel **108** to return to a straight condition where the axle height **172** is lowest. When the front wheel **108** becomes unloaded after impact with a bump or rough terrain and happens to turn to one side or the other, further contact with the ground encourages the front wheel **108** to straighten out. In this way, the negative rake angle **164** of the head tube **130** contributes to a more stable ride of the powered two-wheel vehicle **100**.

[0061] At higher speeds, such an encounter with a bump or crack in the terrain and rocks or other foreign objects can induce caster flutter in the steering assembly **128**, even though the steering assembly **128** is generally resistant to a caster flutter condition. The inclusion of the rotational damper **156** in the steering assembly enhances the stability already provided by the geometry of the steering assembly **128** by, for example, prohibiting caster flutter at high speeds. The rotational damper **156** connected between the head tube **130** and the steering tube **132** provides a dampening effect—i.e., resistance to movement—on the rotation of the steering tube **132**. A wide variety of mechanisms can be used to resist rotational movement of the steering tube **132** relative to the head tube **130**, some of which are described in greater detail below. In some powered two-wheel vehicles, the dampening effect increases in proportion to the angular velocity of the steering tube **132**; put another way, the rotational damper **156** can resist rotation the faster the steering tube **132** rotates. Consequently, when the rider leans in a direction to turn the powered two-wheel vehicle **100**, the front wheel **108** is prohibited from jerking too quickly in either direction. Similarly, when the rider hits a bump or rough terrain that causes the front wheel **108** to become unloaded and even lift off of the ground, the front wheel **108** is prohibited from turning too quickly such that re-engagement of the front wheel **108** with the ground causes a sudden turn. Thus, inadvertent, or unintended turning and caster flutter at high speeds are prohibited and the stability of the powered two-wheel vehicle **100** is maintained.

[0062] As is noted above, the rotational damper **156** can take on a wide variety of forms, such as, for example, a spring damper, a linear-piston damper, a polymer-based friction damper, a rotational disc damper, a friction damper, a viscoelastic damper, a hydraulic damper, and the like. In some powered two-wheel vehicles, the dampening effect is provided by increasing friction between the steering tube **132** and the head tube **130** using a bushing, an O-ring, or other annular component that is held in place and optionally compressed by a nut. As noted above, these mechanisms resist the rotation of the steering tube **132** relative to the head tube **130** and other components of the steering assembly **128**. The resistance force applied by the rotational damper **156** can be uniform or can change in proportion to the rotation of the steering tube **132**. Some rotational dampers apply a resistance force that increases in proportion to the magnitude of the rotation, that is, how far the steering tube **132** has turned from the neutral or straight position. Some rotational dampers apply a resistance force that increases in proportion to the angular velocity of the rotation, that is, how quickly the steering tube **132** is turning. Some rotational dampers apply a resistance force that increases with both magnitude and angular velocity of rotation of the steering tube **132**. Still other rotational dampers apply a resistance force that is designed to increase and/or decrease according to the magnitude and/or angular velocity of rotation of the steering tube **132** according.

[0063] The maximum amount that the front wheel **108** can turn is limited by a center stop **174** extending in a rearward direction from the head tube **130** and two side stops **176** that extend upward from the front fork **104**. As the front fork **104** and front wheel **108** rotate, the side stops **176** engage the center stop **174** to prohibit rotation of the front fork **104** beyond a predetermined rotation angle measured from a centerline of the powered two-wheel vehicle **100** to the side stop **176**, such as, for example, from 15 degrees to 40 degrees, from 20 degrees to 35 degrees, or from 25 degrees to 30 degrees.

[0064] The powered two-wheel vehicle **100** described herein can also be provided as a rolling chassis, i.e., including all of the components of the powered two-wheel vehicle **100** except for the battery **138** and the control system **142** that are provided by the purchaser of the rolling chassis. A purchaser of the rolling chassis version of the vehicle **100** can supply their own battery and control system to facilitate the operation of the powered two-wheel vehicle **100**. In this way, the purchaser can take advantage of all of the benefits of the steering assembly **128** and the motor **136** that provide a stable ride while adding their own electronic systems that they can experiment with.

[0065] Referring now to FIGS. **11** and **12**, the exemplary powered two-wheel vehicle **100** is shown again with optional features such as, for example, a front fender **180**, a rear fender **182**, a front light bracket **198**, and a rear light bracket **200**. In some embodiments, the front fender **180** can be attached to and removed from the frame tubes **134** without the use of tools. Referring now to FIGS. **13-20**, the front fender **180** and a method of assembling the front fender **180** to the powered two-wheel vehicle **100** without tools are shown. Attaching the front fender **180** to the frame tubes **134** obstructs the path of water and other road debris that might otherwise splash up between the frame tubes **134** and onto the rider.

[0066] The front fender **180** has a curved shape that conforms to the shape of and attaches to the frame tubes **134**. The front fender **180** includes flexible side walls **184** that partially wrap around the frame tubes **134**. A protrusion **186** extending downward between the flexible side walls **184** facilitates the alignment of the front fender **180** between the frame tubes **134**. The front fender **180** is shaped to pass between the frame tubes **134** and engage the front bulkhead **118**. The front fender **180** narrows to form side recesses or cut-outs **188** that accommodate the frame tubes **134** as the front fender **180** passes between the frame tubes **134** and widens at a bottom end **190**. The front fender **180** can be shaped so that when the front fender **180** is attached to the frame tubes **134** the bottom end **190** is biased against the front bulkhead **118**. The biasing of the bottom end **190** toward the front bulkhead **118** can be achieved by forming the front fender **180** into a curved shape that has a larger radius than the curve of the frame tubes **134** so that the front fender **180** is partially bent during installation.

[0067] The rear fender **182** is secured to the guard **126** and is supported above the rear wheel **110** by a rear fender support **192** that attaches to the rear fork **106**. Thus, the front fender **180** and the rear fender **182** can be easily removed from and attached to a powered two-wheel vehicle **100** so that fenders **180**, **182** having a different appearance can be attached to the powered two-wheel vehicle **100** based on the preference of the rider. For example, the fenders **180**, **182** can be formed in a wide variety of shapes, in a wide variety of colors, and can be covered in various images and graphics.

[0068] The front fender **180** is shown being attached to the frame tubes **134** in FIGS. **17-20**. As can be seen in FIGS. **17-18**, the front fender **180** is rotated sideways so that the bottom end **190** can fit between the frame tubes **134** even though the bottom end **190** can be wider than the gap between the frame tubes **134**. The front fender **180** is inserted between the frame tubes **134** until the side cut-outs **188** are approximately aligned with the frame tubes **134** and is then rotated in the direction shown by the arrow **194**. The front fender **180** is then rotated until the protrusion **186** is aligned between the frame tubes **134** and the flexible side walls **184** are aligned with the sides of the frame tubes **134**. The front fender **180** is then moved downward toward the frame tubes **134** as indicated by the arrow **196** in FIGS. **19-20**. Pressing the front fender **180** downward with the protrusion **186** between the frame tubes **134** causes the flexible side walls **184** to spread apart and then snap back toward each other once the front fender **180** is fully attached to the frame tubes **134**.

[0069] Referring again to FIGS. **11-12**, the powered two-wheel vehicle **100** can optionally include a front light bracket **198** and/or a rear light bracket **200**. Lights (not shown) can be attached to the front and rear light brackets **198**, **200** to provide illumination of the path ahead of the powered two-wheel vehicle **100** and/or awareness of the presence of the powered two-wheel vehicle **100** to others who are following behind the rider on the powered two-wheel vehicle **100**. The lights can be batter powered or can be powered from the battery **138** of the powered two-wheel vehicle **100** via electrical wires extending from the front or rear lights to the battery **138** or to receptacles that are electrically connected to the battery **138**.

[0070] The powered two-wheel vehicle **100** of FIGS. **11-12** and **17-20** also includes an electro-mechanical rotational damper **204** that can be provided in place of or in addition to the rotational damper **156** described above. The electro-mechanical rotational damper **204** provides a variable damping force based on a damping force signal sent from the control system **142**. The variable damping force applied to the steering tube **132** by the electro-mechanical damper **204** can permit the steering tube **132** to rotate freely and can provide sufficient damping force to prohibit rotation of the steering tube **132**. The damping force exerted by the electro-mechanical damper **204** can be generated by an electric motor, an electric actuator, electromagnets, or the like and can incorporate features of the rotational dampers described above.

[0071] The damping force applied by the electro-mechanical rotational damper **204** on the steering tube **132** can vary based on the rotational position and/or the angular velocity of the steering tube **132** relative to the head tube **130**. The amount of damping force exerted by the electro-mechanical rotational damper **204** can also vary based on the speed of the powered two-wheel vehicle **100** over the ground. That is, the control system **142** can send a different damping force signal to the electro-mechanical damper **204** depending on the current state of the powered two-wheel vehicle **100** and performance parameters established at the factory or by the rider. In some embodiments, the electro-mechanical damper **204** can apply sufficient force to prohibit rotation of the steering tube **132** so that the rotational position of the front wheel **108** is locked until the control system reduces the damping force so that the front wheel **108** is free to turn.

[0072] Referring now to FIG. **21**, a block diagram of the control system **142** is shown to illustrate inputs **206** that can be received by and outputs **208** that can be generated by the control system **142**. The values of the inputs **206** and the outputs **208** can be recorded and stored in a memory unit of the control system **142** or another storage unit for later retrieval and analysis locally or for transmission wirelessly to a server for aggregation and analysis of the data transmitted from a plurality of powered two-wheel vehicles **100**. For example, the owner of the powered two-wheel vehicle **100** may want to download the data during a recent ride for analysis or comparison with other riders. As another example, the owner of a fleet of powered two-wheel vehicles **100** may want to received data transmitted from each vehicle in the fleet to keep track of the vehicle performance and location.

[0073] The inputs or input signals **206** that can be received by the control system **142** include user input **210** from the user input device **146**, location data **212**, altitude data **214**, vehicle performance data **216**, battery data **218**, motor data **220**, steering data **222**, vehicle status data **224**, sensor data **226**, and the like. The user input data **210** can include the selected performance mode, an instruction to accelerate or decelerate, an instruction to increase or decrease steering damping, an instruction to activate or deactivate lights or a horn attached to the powered two-wheel vehicle **100**, an instruction to power the vehicle on or off, or the like. The location data **212** can be provided by a GPS or other geo-location device and can include the geographic location, altitude, velocity, and other GPS-related data related to the powered two-wheel vehicle **100**. Altitude data **214** can be provided alongside the location data **212** from a GPS device or can be provided from an altimeter. The vehicle performance data **216** can be provided by an accelerometer, a gyroscope, or the like and can include the current speed, acceleration or deceleration, and orientation or tilt of the powered two-wheel vehicle **100**. The battery data **218** can include the current, voltage, temperature, and capacity of the battery **138**. The motor data **220** can include the input current, input voltage, temperature, output torque, and output RPM. The steering data **222** can include the angle and/or the angular velocity of the front wheel **108** relative to the powered two-wheeled vehicle **100**. The vehicle status data **224** can include whether the vehicle is powered on, whether optional lights are activated, whether a user input device is connected to the vehicle, and the like. The sensor data **226** can include data from a wide variety of sensors for monitoring properties of the powered two-wheel vehicle **100** and the surrounding environment, such as, for example, cameras for gathering visual data, microphones for gathering audio data, and moisture sensors for detecting water inside the powered two-wheel vehicle **100**.

[0074] The outputs or output signals **208** that can be generated by the control system **142** include signals for motor control **228**, braking control **230**, damper control **232**, light control **234**, speaker control **236**, user feedback **238**, data recording **240**, and the like. Motor control signals **228** can include an instruction to accelerate, to decelerate, and to limit the power output—for example, to limit the maximum speed or acceleration. A braking control signal **230** can instruct mechanical or electro-mechanical breaks to actuate or can be similar to a deceleration signal with an increased rate of deceleration depending on the magnitude of the braking signal. Steering damper control signals **232** can include an instruction to increase or decrease the damping force applied to the steering tube **132** and to apply maximum damping force to lock the steering tube **132** in place. Light control signals **234** can instruct various lights of the powered two-wheel vehicle **100** to activate or deactivate, change intensity, and change color. Speaker control signals **236** can include an instruction to operate a horn and play sounds out of speakers. User feedback signals **238** provide audiovisual and tactile feedback to the rider via displays, speakers, vibration motors, and the like on the powered two-wheel vehicle **100** and similar feedback on the user input device **146** and/or a user's phone, tablet, computer, or the like. User feedback signals can include speed, the selected operating mode, battery level, distance traveled, amount of progress along a route, estimated remaining range, alerts or warnings, and the like. Data recording signals **240** transmit data related to the inputs **206**, outputs **208**, and a wide variety of values computed or derived from the inputs **206** and outputs **208** to a data storage device that can be part of the control system **142** or can be a separate memory storage device for recording the data transmitted from the control system.

[0075] The output signals **208** of the control system **142** can be monitored and fed back into the control system **142** where the input signals **206** and the output signals **208** can be evaluated to facilitate feedback control and machine learning to improve the performance and usability of the powered two-wheel vehicle **100**. For example, the control system **142** can monitor the input signals **206** to determine how well the rider is able to balance and control the powered two-wheel vehicle **100** at various speeds and under different conditions to evaluate whether the selected operating mode is appropriate for the rider's ability and suggest a different operating mode if warranted by the outcome of the evaluation. The input signals **206** and output signals **208** can also be monitored and evaluated to determine the efficiency of the motor **136** over time to better predict the estimated range of the powered two-wheel vehicle **100** and when maintenance or repairs may be necessary. As another example, the inputs **206** from, for example, an accelerometer, a gyroscope, a microphone, and the like can in combination indicate that the rider has been in a crash. Crash data can be taken into consideration when recommending a change in operating mode, can be recorded for a vehicle used as part of a fleet of vehicles, and can be reported to a central server.

[0076] The control system **142** can also store custom performance profiles that alter the performance of the powered two-wheel vehicle **100** during use. For example, a custom acceleration profile or curve can be used by the control system **142** to determine how quickly to accelerate based on the current speed and the operating mode selected by the user. As another example, a custom damping profile can be used by the control system to determine how much damping force to apply to the steering tube **132** via the electro-mechanical damper **204** based on the operating mode selected by the user, the current speed of the vehicle, and the steering data—e.g., the steering angle and angular velocity of the front wheel. The custom profiles can be accessed and edited by the user via a computer, a smart phone, a tablet, or the like using an application or a web interface connected to the powered two-wheel vehicle **100**. The user input device **146** that includes a display **202**, such as the user input device **146** shown in FIG. **11**, can also be used to edit the custom profiles without the use of an additional device. Thus, a user can tweak the performance characteristics of the powered two-wheel vehicle **100** to suit their needs.

[0077] The control system **142** can prohibit operation of the powered two-wheel vehicle **100** absent an authentication device such as a key, a passcode, or a biometric authenticator (e.g., a fingerprint reader). The control system **142** can also store various settings and associate those stored settings with a particular user, such as, for example, custom performance profiles and/or a selected operating mode. Thus, one powered two-wheel vehicle **100** can be shared between multiple users, each with their own personalized settings. Recorded data—e.g., ride history, geo tracking, crash history, etc.—can also be associated with the user who had unlocked the powered two-wheel vehicle **100**. When operating a fleet of vehicles, an owner or master account may be able to access all of the data of all users of all of the vehicles to facilitate monitoring of usage patterns by certain users. Associating ride data with a particular user can be used by the control system **142** or by a central server to generate—using standard programming techniques or via machine learning or artificial intelligence—a profile for the user to better tailor the performance of the powered two-wheel vehicle **100** to whichever rider is currently riding the vehicle **100**.

[0078] Referring again to FIGS. **1** and **11**, the user input device **146** is shown in various forms, including with an optional display **202** in FIG. **11**. The display **202** can be used to present outputs **208** from the control system **142** to the user and can also be used to present a user interface to the user so that the user can interact with data stored in the control system **142** or for adjusting parameters of the powered two-wheel vehicle **100**. The user input device **146** is illustrated in an abstract manner and can take on a wide variety of shapes and form factors, for example, to facilitate being easily held by the user. In some embodiments, the user input device **146** and the powered two-wheel vehicle **100** can also interface wirelessly with a smart phone to act as another display **202** or in place of the display **202** on the user input device **146**.

[0079] As was noted above, the buttons **150** on the user input device **146** can be used to select an operating mode of the powered two-wheel vehicle **100**. As was previously noted, the different operating modes can alter the performance characteristics of the powered two-wheel vehicle to be more suitable for a particular terrain or for a particular rider. Examples of operating modes can include power control modes, damping control modes, and combinations thereof. Power control modes can include a low-power mode, a medium-power mode, and a high-power mode. In the low-power mode, the control system **142** limits current provided to the motor **136** to limit the maximum acceleration of the powered two-wheel vehicle **100** (and consequently, the top speed) which may be more suitable to novice riders. Limiting maximum acceleration also conserves the capacity of the battery **138** to extend the range of the powered two-wheel vehicle **100**. In the high-power mode, the control system **142** makes higher current available to the motor **136**—in some embodiments, the maximum rated current for the battery **138**—for rapid acceleration and higher top speed for the powered two-wheel vehicle **100** that is suitable for more experienced riders. In high-powered mode, the range of the powered two-wheel vehicle **100** can be significantly limited because of the high current drain on the battery **138** to provide increased acceleration to the motor **136**. The mid-power mode strikes a balance between the low-power and the high-power modes, limiting current to the motor **136** more than the high-powered mode so that moderate acceleration is available to the powered two-wheel vehicle **100** without significantly reducing the estimated range. The mid-power mode is the default operating mode for the powered two-wheel vehicle **100**.

[0080] The damping control modes can include a fixed damping mode, a speed-based damping mode, and a user-input or freestyle damping mode that each provide different damping performance of the steering assembly **128** that includes an electro-mechanical rotational damper **204**. Any of these damping control modes can be combined with any of the power control modes described above. The buttons **160** on the user input device **146** can be used to select the damping control mode separate from the power control mode, or in conjunction with the power control mode. In fixed damping mode, the damping force provided by the electro-mechanical rotational damper **204** is set to a fixed amount that can be adjusted by the user, similar to a rotational disc friction damper. In the speed-based damping mode, the damping force provided by the electro-mechanical damper **204** increases with the speed of the powered two-wheel vehicle **100**. The damping force can increase in a linear relationship to the vehicle speed according to a linear damping profile so that steering is looser and more maneuverable at low speeds and stiff at higher speeds. The damping force can also increase according to a custom damping profile selected and/or edited by the user. In the user-input or manual damping mode, the damping force provided by the electro-mechanical damper **204** varies according to input provided by the user via the user input device **146**. The user input device **146** can include a trigger-style input **210** that allows the user to control the amount of damping force. Fully engaging the trigger-style input **210** provides maximum damping force to the steering assembly **128** to lock the position of the front wheel **108**. The manual damping mode enables a user to increase damping when cornering and to lock the front wheel **108** position entire when performing freestyle tricks.

[0081] Referring now to FIG. **22**, a methodology **300** that facilitates the control of the powered two-wheel vehicle **100** is shown. The methodology **300** can be performed with any of the powered two-wheel vehicles described herein. The methodology **300** begins at **302** with receiving a user input signal **210** from a user input device **146** at a control system **142**. The user input signal **210** includes an acceleration instruction or a deceleration instruction. The control system **142** also receives, at **304**, a selected operating mode and at least one of vehicle performance data **216**, motor data **220**, steering data **222**, vehicle status data **224**, and sensor data **226**. At **306**, the control system **142** generates a motor control signal **228** based on: the user input signal **210**; at least one of the vehicle performance data **216**, motor data **220**, steering data **222**, vehicle status data **224**, and sensor data **226**; and the selected operating mode. The motor **136** is operated in response to or according to the motor control signal **228** at **308**, for example, to accelerate or to decelerate the powered two-wheel vehicle **100**. As is described above, the selected operating mode can be a power control mode that includes a low-power mode, a medium-power mode, and a high-power mode. The steps of controlling the powered two-wheel vehicle **100** can be performed sequentially or simultaneously.

[0082] In some exemplary methodologies, the user input signal includes user authentication information, and the methodology includes steps of verifying whether the user is permitted to operate the powered two-wheel vehicle based on the user authentication information, and permitting operation of the powered two-wheel vehicle based on a verification that the user is permitted to operate the powered two-wheel vehicle. The methodology can also include steps of generating a data recording signal, storing the data recording signal in a data storage device, and associating the stored data recording signal with the user of the powered two-wheel vehicle.

[0083] Examples of powered two-wheel vehicles and methodologies for controlling the same are described herein.

[0084] An example of a powered two-wheel vehicle includes a chassis, a deck, a front wheel, a rear wheel, and a steering assembly. The chassis extends between a rotatable front fork and a fixed rear fork. The deck is removably attached to the chassis, wherein attaching the deck to the chassis encloses a compartment. The front wheel is rotatably attached to the rotatable front fork. The rear wheel is rotatably attached to the fixed rear fork via a hub motor, wherein the hub motor provides motive power to the rear wheel. The steering assembly includes a head tube, a steering tube, and a rotational damper. The head tube is attached to the chassis and supported above and in front of the deck by a pair of arched frame tubes, wherein the head tube is tilted forward at a rake angle. The steering tube is rotatably connected to the head tube, wherein the steering tube extends into the head tube from the rotatable front fork so that the rotatable front fork trails the head tube when the powered two-wheel vehicle moves in a forward direction and the front wheel is in contact with a ground surface. The rotational damper is connected between the head tube and the steering tube.

[0085] In an example of a powered two-wheel vehicle, the chassis has two side beams extending between a front bulkhead and a rear bulkhead, and a skid plate extending between the two side beams.

[0086] In another example of a powered two-wheel vehicle, the compartment is enclosed by the deck, the two side beams, the front bulkhead, the rear bulkhead, and the skid plate.

[0087] In yet another example of a powered two-wheel vehicle, the compartment is water-tight.

[0088] In an example of a powered two-wheel vehicle, the rotational damper is a rotational disc damper, and a damping force applied to the steering tube increases in proportion to an angular velocity of the steering tube.

[0089] In another example of a powered two-wheel vehicle, the rotational damper is an electro-mechanical damper.

[0090] In still another example of a powered two-wheel vehicle, the front wheel and the rear wheel each have a diameter in a range of 5 inches to 17 inches.

[0091] In an additional example of a powered two-wheel vehicle, a height of the deck is below a height of a center of each of the front wheel and the rear wheel.

[0092] Another powered two-wheel vehicle includes a battery and a control system disposed in the compartment. The battery provides electrical power to the hub motor and the control system. The powered two-wheel vehicle also includes a user input device for transmitting user input signals to the control system. The user input device includes a plurality of control elements for controlling the operation of the powered two-wheel vehicle.

[0093] In an example of a powered two-wheel vehicle, actuating one of the plurality of control elements of the user input device in a first direction sends an acceleration instruction to the control system and actuating the control element in a second direction sends a deceleration instruction to the control system. The control system generates a motor control signal based on the acceleration instruction or deceleration instruction and an operating mode of the control system.

[0094] In another example of a powered two-wheel vehicle, the operating mode is a power control mode including a low-power mode, a medium-power mode, and a high-power mode.

[0095] In an example of a powered two-wheel vehicle, the rotational damper is an electro-mechanical damper and actuating one of the plurality of control elements of the user input device transmits a damping instruction to the control system. The control system generates a damper control signal based on the damping instruction and an operating mode of the control system.

[0096] In another example of a powered two-wheel vehicle, the operating mode is a damping control mode including a fixed damping mode, a speed-based damping mode, and a freestyle damping mode.

[0097] In still another example of a powered two-wheel vehicle, when the speed-based damping mode is selected, the damper control signal is generated based on at least one of a linear damping profile and a custom damping profile.

[0098] In an example of a powered two-wheel vehicle, the rotational damper is a rotational disc damper. A damping force applied to the steering tube increases in proportion to an angular velocity of the steering tube.

[0099] In another example of a powered two-wheel vehicle, in response to a deceleration instruction, the motor control signal generated by the control system causes the motor to slow the powered two-wheel vehicle and to charge the battery.

[0100] In an example of a powered two-wheel vehicle, actuating one of the plurality of control elements of the user input device changes an operating mode of the powered two-wheel vehicle.

[0101] An example method of controlling a powered two-wheel vehicle includes steps of: receiving a user input signal from a user input device at a control system, receiving a selected operating mode and vehicle performance data at the control system, generating a motor control signal, and operating a motor based on the motor control signal. The user input signal includes an acceleration instruction or a deceleration instruction. The motor control signal is generated based on the user input signal, the vehicle performance data, and the selected operating mode.

[0102] In another example of a method of controlling a powered two-wheel vehicle, the selected operating mode can be a power control mode that includes a low-power mode, a medium-power mode, and a high-power mode.

[0103] As another example of a method of controlling a powered two-wheel vehicle includes steps of: receiving a user input signal from a user input device at a control system; receiving at least one of vehicle performance data, motor data, steering data, vehicle status data, and sensor data at the control system; generating a damping control signal; and adjusting a damping force applied to a steering tube of a steering assembly based on the damping control signal generated by the control system. The damping control signal is generated by the control system based on: the user input signal; at least one of the vehicle performance data, motor data, steering data, vehicle status data, and sensor data; and a custom performance profile.

[0104] In another example of a method of controlling a powered two-wheel vehicle, the user input signal comprises user authentication information and the methodology includes steps of verifying whether the user is permitted to operate the powered two-wheel vehicle based on the user authentication information and permitting operation of the powered two-wheel vehicle based on a verification that the user is permitted to operate the powered two-wheel vehicle.

[0105] In still another example of a method of controlling a powered two-wheel vehicle, the method includes steps of: generating a data recording signal; storing the data recording signal in a data storage device; and associating the stored data recording signal with the user of the powered two-wheel vehicle.

[0106] In an example of a method of controlling a powered two-wheel vehicle, generating the damping control signal is also based on an operating mode of the control system, and the operating mode is at least one of a power control mode and a damping control mode.

[0107] In another example of a method of controlling a powered two-wheel vehicle, the power control mode can be a low-power mode, a medium-power mode, and a high-power mode.

[0108] In yet another example of a method of controlling a powered two-wheel vehicle, the damping control mode is a fixed damping mode, a speed-controlled damping mode, and a user-input damping mode.

[0109] Various functions described herein can be implemented in hardware, software, or any combination thereof. If implemented in software, the functions can be stored on or transmitted over as one or more instructions or code on a computer-readable medium. Computer-readable media includes computer-readable storage media. A computer-readable storage media can be any available storage media that can be accessed by a computer. By way of example, and not limitation, such computer-readable storage media can comprise RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to carry or store desired program code in the form of instructions or data structures and that can be accessed by a computer. Further, a propagated signal is not included within the scope of computer-readable storage media. Computer-readable media also includes communication media including any medium that facilitates transfer of a computer program from one place to another. A connection, for instance, can be a communication medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio and microwave are included in the definition of communication medium. Combinations of the above should also be included within the scope of computer-readable media.

[0110] Alternatively, or in addition, the functionality described herein can be performed, at least in part, by one or more hardware logic components. For example, and without limitation, illustrative types of hardware logic components that can be used include Field-programmable Gate Arrays (FPGAs), Application-specific Integrated Circuits (ASICs), Application-specific Standard Products (ASSPs), System-on-a-chip systems (SOCs), Complex Programmable Logic Devices (CPLDs), etc.

[0111] What has been described above includes examples of one or more embodiments. It is, of course, not possible to describe every conceivable modification and alteration of the above devices or methodologies for purposes of describing the aforementioned aspects, but one of ordinary skill in the art can recognize that many further modifications and permutations of various aspects are possible. Accordingly, the described aspects are intended to embrace all such alterations, modifications, and variations that fall within the spirit and scope of the appended claims. Furthermore, to the extent that the term "includes" is used in either the detailed description or the claims, such term is intended to be inclusive in a manner similar to the term "comprising" as "comprising" is interpreted when employed as a transitional word in a claim.

[0112] While various aspects, features and concepts may be expressly identified herein as being inventive or forming part of a disclosure, such identification is not intended to be exclusive, but rather there may be inventive aspects, concepts, and features that are fully described herein without being expressly identified as such or as part of a specific disclosure, the disclosures instead being set forth in the appended claims. Descriptions of exemplary methods or processes are not limited to inclusion of all steps as being required in all cases, nor is the order that the steps are presented to be construed as required or necessary unless expressly so stated. The words used in the claims have their full ordinary meanings and are not limited in any way by the description of the embodiments in the specification.

1-20. (canceled)

21. A powered two-wheel vehicle comprising:
- a chassis extending between a rotatable front fork and a rear fork;
- a front wheel rotatably attached to the rotatable front fork;
- a rear wheel rotatably attached to the rear fork;
- a motor that provides motive power to at least one of the front wheel and the rear wheel; and
- a steering assembly comprising:
  - a head tube attached to the chassis and supported above and in front of the deck; and
  - a rotational damper connected between the head tube and the rotatable front fork.

22. The powered two-wheel vehicle of claim **21**, wherein the rotational damper is a rotational disc damper, and wherein a damping force applied to the steering tube increases in proportion to an angular velocity of the steering tube.

23. The powered two-wheel vehicle of claim **21**, wherein the rotational damper is rotational disc damper.

24. The powered two-wheel vehicle of claim **21**, wherein a height of the deck is below a height of a center of each of the front wheel and the rear wheel.

25. A powered two-wheel vehicle comprising:
- a chassis extending between a rotatable front fork and a rear fork, the chassis comprising a compartment;
- a front wheel rotatably attached to the rotatable front fork;
- a rear wheel rotatably attached to the rear fork;
- a motor that provides motive power to at least one of the front wheel and the rear wheel;
- a steering assembly comprising:
  - a head tube attached to the chassis and supported above and in front of the deck; and
  - a rotational damper connected between the head tube and the rotatable front fork;
- a battery and a control system arranged in the compartment, wherein the battery provides electrical power to the motor and to the control system; and
- a user input device for transmitting user input signals to the control system, wherein the user input device comprises a plurality of control elements for controlling the operation of the powered two-wheel vehicle.

26. The powered two-wheel vehicle of claim **25**, wherein actuating one of the plurality of control elements of the user input device in a first direction sends an acceleration instruction to the control system and actuating the control element in a second direction sends a deceleration instruction to the control system, and wherein the control system generates a motor control signal based on the acceleration instruction or the deceleration instruction and an operating mode of the control system.

27. The powered two-wheel vehicle of claim **26**, wherein motor control signal generated by the control system is also based on vehicle performance data received by the control system.

28. The powered two-wheel vehicle of claim **27**, wherein the wherein the vehicle performance data comprises a current speed of the powered two-wheel vehicle.

29. The powered two-wheel vehicle of claim **25**, wherein the operating mode is a power control mode comprising a low-power mode, a medium-power mode, and a high-power mode.

30. The powered two-wheel vehicle of claim **25**, wherein the operating mode of the control system is selected via the user input device.

31. The powered two-wheel vehicle of claim **25**, wherein the rotational damper is a rotational disc damper, and wherein a damping force applied to the steering tube increases in proportion to an angular velocity of the rotatable front fork.

32. The powered two-wheel vehicle of claim **26**, wherein in response to a deceleration instruction, the motor control signal generated by the control system causes the motor to slow the powered two-wheel vehicle and to charge the battery.

33. The powered two-wheel vehicle of claim **25**, wherein actuating one of the plurality of control elements of the user input device changes an operating mode of the powered two-wheel vehicle.

34. A powered two-wheel vehicle comprising:
- a chassis extending between a rotatable front fork and a rear fork;
- a front wheel rotatably attached to the rotatable front fork;
- a rear wheel rotatably attached to the rear fork;
- a motor that provides motive power to at least one of the front wheel and the rear wheel;
- a steering assembly comprising a rotational damper connected between the chassis and the rotatable front fork;
- a battery that provides electrical power to the motor; and
- a control system configured to:
  - receive at least one of vehicle performance data, a battery data signal, a tilt signal from a gyroscope, an acceleration signal from an accelerometer, and a user input signal; and

generate at least one of a motor control signal, a light control signal, and a user feedback signal.

35. The powered two-wheel vehicle of claim **34**, wherein the motor control signal is generated by the control system based on a current speed of the vehicle performance data and the tilt signal received from the gyroscope.

36. The powered two-wheel vehicle of claim **34**, wherein the light control signal is generated by the control system based on a current speed of the vehicle performance data and the tilt signal received from the gyroscope.

37. The powered two-wheel vehicle of claim **34**, wherein the light control signal is generated by the control system based on a current speed of the vehicle performance data and the acceleration signal received from the accelerometer.

38. The powered two-wheel vehicle of claim **34**, wherein the motor control signal is generated by the control system based on the user input signal and the battery data signal.

39. The powered two-wheel vehicle of claim **38**, wherein the user feedback signal is generated by the control system based on a voltage and a current from the battery data signal.

40. The powered two-wheel vehicle of claim **39**, wherein the user feedback signal is received at a user input device to provide a user with a prompt to select a low-power mode of the powered two-wheel vehicle.

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